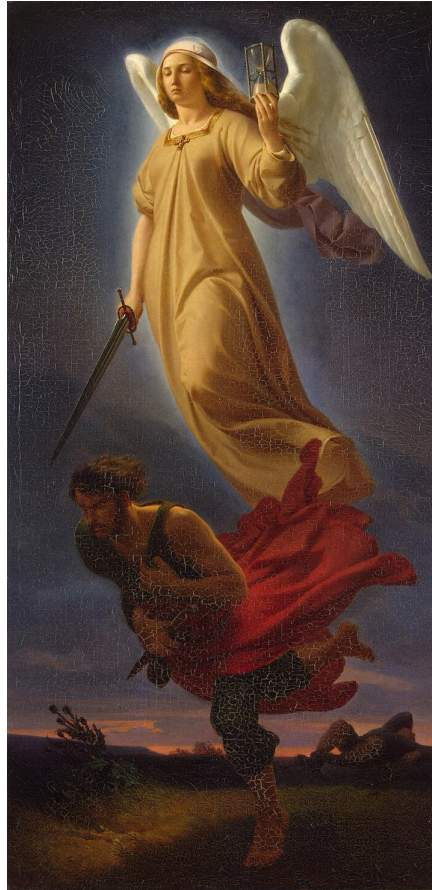


NEMESIS



‘Nemesis’ by Alfred Rethel (1816 – 1859)

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Version 3.0

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0. Overview

This document describes the creation, properties and running of the retrieval model Nemesis – Non-Linear Optimal Estimator for Multivariate Spectral Analysis, which was developed from the Oxcirsg code.

0.1. Reference Documents

- [R1] Irwin, P.G.J, Gradient version of Oxford Radiative Transfer and retrieval Code, Oxford CIRS Technical Report: CIRS/OX/TR/1390.
- [R2] Irwin, P.G.J, and S.B. Calcutt, RADTRAN, Oxford Planetary Technical Report: NIMS/OX/PGJI/SW/136.
- [R3] Hanel R.A., B.J. Conrath, D.E. Jennings and R.E. Samuelson. Exploration of the Solar System by Infrared Remote Sensing: Second Edition, Cambridge University Press, 2003
- [R4] Rodgers, C.D. Inverse methods for atmospheric sounding. Theory and practice. World Scientific. 2000

0.2. Defined Fonts

As in [R1], in an attempt to make this document easier to read the following fonts are used to denote different objects.

- Executable programs are underlined. e.g., CIRSDrvg
- Suites of codes in their own subdirectories are in copperplate font. e.g., **RADTRANS**
- Subroutine files are in courier font. e.g., `cirsradg.f`
- Variables defined within FORTRAN codes are capitalized. e.g., NCONV, IMOD

0.3. Radiance Units

NEMESIS can operate in either wavelength or wavenumber space. Nemesis (or one of its variants) can also compute 5 different types of spectra, defined by the IFORM integer:

IFORM=0	Radiance
IFORM=1	$F_{\text{plan}}/F_{\text{star}}$ i.e., secondary transit depth
IFORM=2	$A_{\text{plan}}/A_{\text{star}}$ i.e., primary transit depth
IFORM=3	Integrated spectral power of planet
IFORM=4	Atmospheric transmission multiplied by solar flux

The actual units used in the .spx files (and .mre files) are:

IFORM=0

- Wavelength Space: $\text{W cm}^{-2} \text{sr}^{-1} \mu\text{m}^{-1}$
- Wavenumber Space: $\text{W cm}^{-2} \text{sr}^{-1} (\text{cm}^{-1})^{-1}$

For the .mre files, these are modified for historical reasons to $\mu\text{W cm}^{-2} \text{sr}^{-1} \mu\text{m}^{-1}$ and $\text{nW cm}^{-2} \text{sr}^{-1} (\text{cm}^{-1})^{-1}$, respectively.

IFORM=1

- Wavelength or Wavenumber Space: $F_{\text{plan}}/F_{\text{star}}$ (dimensionless)

IFORM=2

- Wavelength or Wavenumber Space: $100 \cdot A_{\text{plan}}/A_{\text{star}}$ (dimensionless)

IFORM=3

- Wavelength Space: $W \mu\text{m}^{-1}$
- Wavenumber Space: $W (\text{cm}^{-1})^{-1}$
(Note that these values are scaled internally by a factor of 10^{-18} to ensure numerical stability, but that the final output spectrum in the .mre file has the same units as the input .spx file.)

IFORM=4

- Wavelength Space: $W \text{cm}^{-2} \mu\text{m}^{-1}$
- Wavenumber Space: $W \text{cm}^{-2} (\text{cm}^{-1})^{-1}$

0.4. NEMESIS variants

The standard **NEMESIS** program is called Nemesis, but there are a number of variants, which have had to be defined due to the way they use a different layer scheme, or indeed have different output units. These variants are:

<u>Nemesis</u>	Standard model for modelling individual observations on a planet (IFORM = 0, 1, 3 or 4; Default is IFORM=0). If a .cel file is present the model assumes a Selective Chopper Radiometer simulation.
<u>NemesisL</u>	As <u>Nemesis</u> , but optimised to deal with limb-observing geometries. Model uses different method of combining individual layers to make the calculations faster. (IFORM = 0)
<u>NemesisMCS</u>	Extension of <u>NemesisL</u> to model MCS observations of Mars. Model uses additional FOV data to model observations and also pointing data. (IFORM = 0)
<u>Nemesisdisc</u>	Version of Nemesis for specifically modelling power spectra of planets or secondary transit observations. (i.e., IFORM=1 or 3; Default is IFORM=1). Uses analytical calculation of radiation into a hemisphere and so is only good for non-scattering cases.
<u>NemesisPT</u>	Version of Nemesis for specifically modelling the primary transit spectra of exoplanets. (IFORM = 2). Note that for this model, the functional derivatives may be calculated either implicitly or numerically through the INUMERIC flag (See section 3.2). The numeric differentiation scheme is found to better capture rate of change of primary transit signal with temperature (through temperature's effect on the scale height), but is much, much slower.

1. Introduction

Multivariate retrievals from Composite Infrared Spectrometer (CIRS) spectra, onboard NASA's Cassini mission to the Saturnian system, was the core science objective of the Oxford CIRS data analysis effort. In order to improve the speed of retrievals, the main **CIRSRAD** forward model was overhauled to generate **CIRSRADG**, which calculates the partial derivatives of the synthetic spectra with respect to atmospheric properties internally, instead of calculating these numerically afterwards by calculating numerous 'perturbed' spectra and taking the difference [R1]. Hence, the main **OXCIRS** retrieval code was superseded by **OXCIRSG**, which used this 'gradient' version of the forward model to calculate the Jacobian, or **K**-matrix and was thus much faster.

Following the Cassini spacecraft Jupiter fly-by it was decided to radically overhaul the Oxford CIRS retrieval code to generate a new, general purpose retrieval code which could make maximum use of the advantages of **CIRSRADG** and also be easily switchable between different planets and different observation geometries. The resulting code, Nemesis (Non-Linear Optimal Estimator for Multivariate Spectral Analysis) and has the following advantages over previous codes.

1. **NEMESIS** allows the retrieval of continuous profiles of gas abundance, temperature and cloud. Before, gas and cloud profiles were parameterised and hence the functional derivative capability of Cirsradg was being under-utilised.
2. The layering scheme and state vector elements not hard-wired for a particular planet. Instead, the profile levels used by **NEMESIS** internally are set to the .ref reference profile. The variable elements are then the NVMR gas profiles in the .ref file, the temperature, any of the NCONT aerosol profiles defined, the para-H₂ fraction and the fractional cloud cover. Up to four variable profiles and one other variable (such as surface temperature, surface albedo spectrum or tangent height correction) may currently be simultaneously retrieved (five variables in all).
3. **NEMESIS** allows either nadir or limb observations. Previous codes were unable to simulate limb-viewing conditions.
4. **NEMESIS** models the additional thermal emission from the ground for planets with solid surfaces.
5. **NEMESIS** allows the simultaneous analysis of measurements over a range of observation angles, including combinations of near-nadir and limb views.
6. **NEMESIS** may perform field-of-view averaging.
7. **NEMESIS** may operate in either wavenumber, or wavelength space.
8. **NEMESIS** may also be used for scattering calculations, although in this case the functional derivatives have to be calculated numerically, and thus slowly.
9. **NEMESIS** may perform channel integration, either by numerically convolving a spectrum with a channel filter function, or by the use of channel-integrated k-tables calculated with Calc_fnktablec.
10. **NEMESIS** may now calculate spectra using either the line-by-line (LBL) method in addition to the original correlated-k approximation. Implemented by setting ILBL=1.

11. **NEMESIS** has been extended to be able to model primary and secondary transit spectra of exoplanets.
12. **NEMESIS** has been extended to be able to deal with profiles where the sum of vmrs at each level is made to add up to 1.0. This also means that the molecular weight can be calculated at each level rather than assuming the same value at all levels.
13. **NEMESIS** has been extended to deal with SCR simulations if it detects the presence of a .cel file.
14. **NEMESIS** has been extended to incorporate pre-tabulated monochromatic gas absorption lbl-tables. The tables are generated with Calc_lbltable and are used in Nemesis by setting ILBL=2.

1.1 Why is it called Nemesis?

Nemesis is traditionally known as the Goddess of Vengeance and Retribution. Some authors connect the name with “to feel just resentment” or “righteous anger”. However, the word Nemesis originally meant the distributor of fortune, whether good or bad, in due proportion to each man according to his deserts.

In Greek mythology Nemesis was the daughter of Nyx the primordial goddess of the night. Nyx was born of Chaos. She gave birth to Aether alone and Hemera, Moros, Charon, Eros and the Keres with her brother, Erebus. With Dionysus, she mothered Phthonus. Apart from Nemesis, Nyx was also mother of Momus, Thanatos, Hypnos, the Hesperides, Apate, Philotes, and Geras - the Fates.

Nemesis is said to have been as beautiful as Aphrodite and was seduced by Zeus in the form of a swan. The Goddess of Punishment, Poena, was an attendant of Nemesis.

As Nemesis/Fortuna, a conflation of the Greek deity of fate with the Roman Fortuna, she was perceived not as bringer of retribution, but as having the power of changing fortune. Hence, she was an ideal deity to make patron goddess of gladiators. It is thought that gladiators made offerings to this “goddess of fortune” before fighting in the Roman arenas. It is the “goddess of fortune” view of Nemesis, which inspired the naming of this retrieval code in her honour. It is hoped that Nemesis will bring good fortune and will considerably improve the retrieval of atmospheric properties from remotely-sensed infrared planetary spectra.

2. Running Nemesis

2.1 Input files and running NEMESIS

To run **NEMESIS**, you need some or all (depending on which mode of **NEMESIS** you are using) of the following input files (whose formats are presented in Section 3):

<runname>.inp	File containing details of model run – formerly read from the standard input.
<runname>.nam	File containing name of run. i.e., <runname>
<runname>.set	Contains details of the scattering angles to be used and also how the atmosphere is to be split into layers.
<runname>.ref	Reference atmospheric profile, containing T, P, and gas abundance profiles.
<runname>.cia	File containing the name of the CIA table to be used together with some characteristics of the table.
<runname>.fla	Flag file, containing a list of flag integers that used to be hard-wired into the code.
<runname>.zen	Zenith angle. Code looks to see if this file is present and if so reads in the IPZEN integer which defines the level in the atmosphere at which the zenith angle of the observation is defined [0) at bottom of deepest layer; 1) at the 0km altitude level; or 2) at the very top of the atmosphere]. If the file is not present IPZEN defaults to 0.
aerosol.ref	Reference aerosol profile file.
parah2.ref	Reference para-H ₂ fraction profile. <u>N.B.</u> This file only needs to be present if the planet under consideration is a giant planet and calculation is in wavenumbers. If calculation is in wavelengths then it is assumed that it's in the near-IR in which case there is not yet a tabulation of how CIA varies with para-H ₂ fraction.
fcloud.ref	Reference fractional cloud cover profile. <u>N.B.</u> This file only needs to be present if a scattering run is being undertaken.
<runname>.xsc	Aerosol cross-section file (x-sections and single scattering albedos as a function of wavenumber or wavelength).

- <runname>.sur Surface emissivity file (surface emissivity as a function of wavenumber or wavelength). **N.B.** This file needs to be defined only for cases where the observation is not limb, and the planet is not a giant planet. For scattering calculations (ISCAT=1 to 5) the surface albedo is set to 1-emissivity if GALB, defined in the .set file (and subsequently written to the .sca file) is set negative.
- <runname>.apr *A priori* set up file. Defines variables to be set and how they are represented.
- <runname>.kls List of *k*-distribution table (*.kta) files to be used in calculation.
- <runname>.lls List of lbl-table (*.lta) files to be used in calculation.
- <runname>.spx Spectrum or spectra to be simulated and FOV averaging geometries and weights to use.
- <runname>.abo Abort file, terminates retrievals early if required.
- <runname>.fil Contains channel filter functions. **N.B.** This file only needs to be present if channel integrations are required. See section 3.3.
- <runname>.lbl Contains the wavenumber range, step, wing, v_rel and v_cutoff of lines to be included for LBL Nemesis runs (ILBL=1). **N.B.** only necessary for LBL calculations. Note: the wavenumber range only needs to cover the output spectral range (in wavenumbers of wavelengths) and **does not** additionally need to account for v_rel.
- <runname>.sha Instrument lineshape to be used in final spectral convolution for LBL Nemesis runs (ILBL=1,2). 0=square, 1=triangular, 2=Gaussian, 3=Hamming, 4=Hanning, 5=arbitrary instrument lineshape defined in <runname>.fin. **N.B.** only necessary for LBL calculations.
- <runname>.fin Instrument lineshape file. First line=number of points in linshape, followed on subsequent lines by two columns containing wavenumber/wavelength and instrument lineshape. Optional file. **N.B.** only necessary for LBL calculations with <runname>.sha = 5 option.
- <runname>.key Line data .key file to be used for Nemesis LBL calculations. **N.B.** only necessary for LBL calculations.

- <runname>.pra If present, this file lists which lineshape should be used for which gas absorption lines. **N.B.** only necessary for LBL calculations.
- <runname>.sol The name of the reference solar/stellar spectral that should be used if this is a reflectivity or a transit (either primary or secondary) spectroscopy calculation.
- <runname>.rfl If present, this file has details of any reflecting layer calculations to be added to the output of Nemesis or Nemesisdisc. Described in Section 3.9.
- <runname>.vpf If present, this file lists the gases whose VMRs are to be limited by saturation and for each gas lists the desired limiting relative humidity and whether the volatile is arriving from the deep interior or from space. Described in Section 3.10.
- <runname>.cel If present, this file lists the cell definition parameters that are to be pasted into the .pat file to simulate a Selective Chopper Radiometer (SCR) observation. The format of this section is described in the Radtrans manual. Only SCR cell types are currently supported. If Nemesis detects the presence of this file an SCR simulation is performed.
- <runname>.rdw If present, this file lists the ‘ranking’ of each of the wavelengths in the .spx file, so that the first few iterations of the retrieval model only fits the highest-ranked wavelengths, with the rest of the wavelengths to be fit only in later iterations. This can save a substantial amount of computational time for a retrieval using a large number of wavelengths, while losing almost no information in the final state vector. Note that this scheme cannot yet be used for SCR observations.
- <runname>.pre Raw fitted state vectors and covariance matrices from a previous retrieval run. These are output to <runname>.raw, which is simply renamed to <runname>.pre.
- <runname>.mie The default particle size distribution for the 444 model is a standard Gamma distribution. However, this can be overridden if NEMESIS finds a <runname>.mie, which has the format:
- ```
pvar
mievar(1),xiscatmie(1),xnormmie(1)
...
mievar(pvar),xiscatmie(pvar),xnormmie(pvar)
```

NEMESIS will override ISCAT and NORM for mode mievar(i) by the accompanying xiscatmie(i) and xnormmie(i).

hgphase(1-n).dat NCONT files in all containing the Henyey-Greenstein phase functions for each particle type as a function of either wavenumber or wavelength. **N.B.** These files are only required for scattering calculations.

The format of some of these files is described in the next section.

The code is then run either by typing 'Nemesis' and then entering <runname>, or by typing 'Nemesis < runname.nam > test.prc &'. Note, that the last part of this line, i.e., '> test.prc &', redirects diagnostic print information to an output file rather than the screen and the '&' part makes the program run in the background. By default, **NEMESIS** outputs a lot of diagnostic information to the screen. This can be suppressed, if the user wishes, but editing the nemesis.f file and setting IDIAG=0. In addition, some of writing to meta files (such as the <runname>.drv) file can be suppressed by additionally setting IQUIET=1 in nemesis.f. Obviously, the code needs to be recompiled to implement these changes.

N.B. There are additional slight variants of **NEMESIS**, described in sections 0.4 and 8, such as **NEMESISL**, and **NEMESISDISC**, which are optimised specific geometries.

## 2.2 Intermediate files

Intermediate files created by **NEMESIS** to run the Cirsradg (or Cirsrad for a scattering calculation) routines, on which it is based, are:

|               |                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------|
| <runname>.pat | Radtrans path file [R2].                                                                                   |
| <runname>.sca | Radtrans scattering file.                                                                                  |
| <runname>.str | meta file for passing previously retrieved parameters to the routine which writes the .prf files.          |
| <runname>.prf | Radtrans atmospheric profile file                                                                          |
| aerosol.prf   | Radtrans aerosol profile file                                                                              |
| parah2.prf    | Radtrans para-H <sub>2</sub> profile file (if planet is a Giant Planet and calculation is in wavenumbers). |
| fcloud.prf    | Radtrans fractional cloud cover profile file (if you are doing a scattering calculation).                  |

Intermediate files created by Cirsradg/Cirsrad:

|               |                       |
|---------------|-----------------------|
| <runname>.drv | Radtrans driver file. |
|---------------|-----------------------|

## 2.3 Output files and inspecting output

The final output file of **NEMESIS** is:

`<runname>.mre` Fitted spectrum, retrieved state vector and errors for one or several measurements (defined in `<runname>.spx`). This file can be plotted by either the IDL routines `plotmretnewX.pro` (for individual profiles) or `scanmretnewX.pro` (for plotting several retrievals consecutively).

There is also an additional output file for version A7 onwards:

`<runname>.raw` Raw fitted state vectors and covariance matrices. These are output in case the results of previous retrievals (including retrieval errors) are required in later retrievals, in which case this file is renamed as `<runname>.pre`.

In addition, if only one profile is retrieved from the input `.spx` file, then a number of other output files are written for diagnostic purposes:

`<runname>.itr` Full ASCII record of the state vector, fitted spectrum and **K**-matrix for each iteration of the retrieval. Can be read and plotted by the IDL routine `plotiternewX.pro`.

`<runname>.cov` ASCII 'Covariance' file. Contains the **K**-matrix as well as the gain matrix **G**, averaging kernels **A** and error covariance matrices. Can be read and displayed by the IDL routine `imagecovariance.pro`.

`kk.out` Unformatted binary file containing the **K**-matrix. Can be plotted with the IDL routine `plotkkimageX.pro`.

## 3. Input File Formats

Most of the input files required are standard Radtrans format files described by [R2]. The `.ref` files are basically `.prf` files and provide reference profiles that remain static during a **NEMESIS** run. The format of the temperature/pressure/vmr `.prf` and `.ref` files has recently been updated: 1) both now list the volume mixing ratios of all the gases on the **same** lines as the height, pressure and temperature, rather than in separate 6-column blocks; and 2) the `.ref` file can now include reference profiles for a range of different latitudes – the number of latitudes included is listed after the AMFORM parameter and the file then holds a set of `.prf` files at these different latitudes. We assume that all the profiles use the same AMFORM and hold the same gas vmrs.

The actual profiles used at each iteration of the forward model are generated from the `.ref` files and the variable profiles defined in the `.apr a priori` file and are written to intermediate `.prf` files. The formats of the other major input files will now be described.

### 3.1 *A priori* .apr file.

The .apr file format is as follows:

---

```
***** any header info you like. One line only *****
NVAR ! number of variable profiles (vmr,T, or cont)
VARIDENT(1,1:3) ! Identity of profile 1
VARPARAM(1,*) ! Any extra parameters, or filename
VARIDENT(2,1:3) ! Identity of profile 2
VARPARAM(2,*) ! Any extra parameters, or filename)
Etc.
```

---

An example .apr file is:

---

```
***** any header info you like. One line only *****
2 ! number of variable profiles (vmr,T, or cont)
11 0 1 ! Ammonia, deep, fsh
0.7 ! pknee
2.19e-4 2.19e-5 ! deep vmr and error
0.15 0.05 ! fsh and error
0 0 0 ! Temperature - continuous
tempapr.dat
```

---

The top line of the file is assumed to contain header information and is skipped. The next line contains NVAR, the number of variable profiles that are to be fitted. For each variable profile, the .apr file contains VARIDENT(IVAR,1:3) which is read in next. In the case above, the VARIDENT(IVAR,1:3) of the first variable is 11, 0, 1. The first two integers describe the identity of the profile, and the third integer describes how the profile is parameterised. The profile may be gas abundance, temperature, aerosol density, para-H<sub>2</sub> fraction, surface temperature, surface albedo spectrum or tangent height correction, etc., depending on VARIDENT(IVAR,1) as follows. **Note that for model types with VARIDENT(IVAR,1) greater than NGAS, then VARIDENT(IVAR,3) should be set equal to VARIDENT(IVAR,1).**

- 1) If VARIDENT(IVAR,1) is greater than 0 then the profile is a gas volume mixing ratio, and the first two integers then contain IDGAS and ISOGAS respectively, as defined by the Radtrans manual [R2].
- 2) If VARIDENT(IVAR,1) is equal to zero, then the profile is a temperature profile.
- 3) If VARIDENT(IVAR,1) is less than zero, then the profile is either aerosol density **OR** para-H<sub>2</sub> fraction **OR** fractional cloud cover. Defining N as -VARIDENT(IVAR,1), if  $N \leq \text{NCONT}$  (the number of aerosol types defined in aerosol.ref and runname.xsc) then the profile is aerosol density with ICONT = N.
- 4) If VARIDENT(IVAR,1) is less than zero and  $-\text{VARIDENT(IVAR,1)} = \text{NCONT} + 1$ , then the profile is the para-H<sub>2</sub> fraction.
- 5) If VARIDENT(IVAR,1) is less than zero and  $-\text{VARIDENT(IVAR,1)} = \text{NCONT} + 2$  then the profile is the fractional cloud cover.
- 6) If VARIDENT(IVAR,1) is equal to 999 then the parameter described is the surface temperature. **Relevant only for non-giant planets.** The next line of the .apr file then contains the *a priori* surface temperature and error.

- 7) If VARIDENT(IVAR,1) is equal to 888 then the parameter described is a surface albedo spectrum. **Relevant only for non-giant planets.** The next line contains the number of wavelengths/wavenumbers for which the surface albedo spectrum is tabulated and the correlation length (in wavelengths). Following lines contain the wavelengths/wavenumbers and the *a priori* albedos and errors. The number of spectral points and the wavelengths/wavenumbers should agree with those defined in the accompanying .sur file.
- 8) If VARIDENT(IVAR,1) is equal to 889 then the parameter described is a surface albedo scaling factor. **Relevant only for non-giant planets.** The next line contains the *a priori* scaling factor and error.
- 9) If VARIDENT(IVAR,1) is equal to 887 then the parameter described is the cross-section spectrum of one of the cloud types. The next line contains the number of wavelengths/wavenumbers for which the surface albedo spectrum is tabulated, the cloud ID (1 – N\_CLOUD) and the correlation length (in wavelengths). Following lines contain the wavelengths/wavenumbers and the *a priori* albedos and errors. The number of spectral points and the wavelengths/wavenumbers should agree with those defined in the accompanying .xsc file.
- 10) If VARIDENT(IVAR,1) is equal to 777 then the parameter described is a correction to the tangent height altitude for limb observations. The next line contains the assumed tangent height correction (in km) together with the error.
- 11) If VARIDENT(IVAR,1) is equal to 666 then the parameter described is a retrieval of the pressure at a defined altitude used for Mars MCS limb observations. The next line contains the assumed defined altitude and the following line gives the assumed pressure together with the error.
- 12) If VARIDENT(IVAR,1) is equal to 555 then the parameter described is a retrieval of the planetary radius. The next line contains the assumed radius (in km) correction together with the error.
- 13) If VARIDENT(IVAR,1) is equal to 449 then the parameter described is a retrieval of the imaginary part of a cloud's complex refractive index spectrum. Model based on 444. This model has three variables: the size distribution mean radius, the variance of the distribution and a scaling factor by which the *a priori*  $n_{\text{imag}}$  spectrum should be multiplied.
- 14) If VARIDENT(IVAR,1) is equal to 448 then the parameter described is a retrieval of the size distribution of particles with known complex refractive index spectrum. Model based on 444. This model has two variables: the size distribution mean radius, and the variance of the distribution.
- 15) If VARIDENT(IVAR,1) is equal to 447 or 446 then the parameter described is a retrieval of the imaginary part of a cloud's complex refractive index spectrum. Model based on 444. However, these models now appear to be vestigial.
- 16) If VARIDENT(IVAR,1) is equal to 445 then the parameter retrieves a cloud's imaginary refractive index using a Mie coated sphere model (Toon and Ackerman, 1981). The cloud particle identifier is given by VARIDENT(IVAR,2). The next line contains the name of a separate input file, which contains the following information. Line 1 contains the mean radius of the particle size distribution and error (assumes standard size distribution), while line 2 gives the variance of the size distribution and error. Line 3 gives the ratio of the radius of the outer 'shell' of a single cloud particle with respect to the total radius of the cloud particle, with error. Line 4 gives the number of wavelengths/wavenumbers for which the imaginary refractive index spectrum to be retrieved is tabulated, together with the correlation length of the corresponding *a priori* spectrum. Line 5 gives a reference

wavelength/wavenumber and the real parts of the refractive indices of the core and shell respectively at that reference. Line 6 gives the wavelength/wavenumber to which the extinction cross-section spectrum should be normalised. Following lines contain the wavelengths/wavenumbers and the *a priori* values of the two imaginary refractive index spectra to be retrieved with errors (core in second and third columns, shell in fourth and fifth). In this model, the real parts of the refractive index spectra are calculated with a Kramers-Kronig analysis and then the Mie scattering properties of the particles calculated using `dmiess.f`. The wavelengths/wavenumbers do not need to match those in the accompanying `<runname>.xsc` file – they will be interpolated over. Note also that if the correlation length is set to be less than 0.0 then the code assumes you want to use the same imaginary refractive index at all wavelengths and just retrieves one parameter. In this case the *a priori* value of  $n_{\text{imag}}$  is taken to be the first one that is listed in the  $n_{\text{imag}}$  spectrum.

- 17) If `VARIDENT(IVAR,1)` is equal to 444 then the parameter described is a retrieval of the imaginary part of a cloud's complex refractive index spectrum. The cloud particle identifier is given by `VARIDENT(IVAR,2)`. The next line contains the name of a separate input file, which contains the following information. Line 1 contains the mean radius of the particle size distribution and error (assumes standard size distribution), while line 2 gives the variance of the size distribution and error. Line 3 gives the number of wavelengths/wavenumbers for which imaginary refractive index spectrum is tabulated, together with the correlation length of this *a priori* spectrum. Line 4 gives a reference wavelength/wavenumber and the value of the real part of the refractive index at that reference. Line 5 gives the wavelength/wavenumber to which the extinction cross-section spectrum should be normalised. Following lines contain the wavelengths/wavenumbers and the *a priori* values of the imaginary refractive index spectrum and errors. In this model, the code the real part of the refractive index spectrum is calculated with a Kramers-Kronig analysis and then the Mie scattering properties of the particles calculated. The wavelengths/wavenumbers do not need to match those in the accompanying `<runname>.xsc` file. Note also that if the correlation length is set to be less than 0.0 then the code assumes you want to use the same imaginary refractive index at all wavelengths and just retrieves one parameter. In this case the *a priori* value of  $n_{\text{imag}}$  is taken to be the first one that is listed in the  $n_{\text{imag}}$  spectrum.
- 18) If `VARIDENT(IVAR,1)` is equal to 443 then the parameter retrieves a cloud with a variable top pressure, deep value and scattering power law index. The scale height is assumed to be equal to the atmospheric pressure scale height. The second line of the parameterization in the `.apr` requires placeholder values one. This parameterization should only be used in MultiNest mode.
- 19) If `VARIDENT(IVAR,1)` is equal to 442 then the parameter retrieves a cloud with a variable top and base pressures, deep value and scattering power law index. The scale height is assumed to be equal to the atmospheric pressure scale height. The second line of the parameterization in the `.apr` requires placeholder values one. This parameterization should only be used in MultiNest mode.
- 20) If `VARIDENT(IVAR,1)` is equal to 441 then the parameter retrieves a cloud based on the parameterization used by MacDonald & Madhusudhan. Below a variable knee pressure a grey, infinitely opaque cloud is included. Above this pressure is a haze with variable opacity and scattering power law index. . The second line of the parameterization in the `.apr` requires placeholder values one. This parameterization should only be used in MultiNest mode.

- 21) If VARIDENT(IVAR,1) is equal to 440 then the parameter retrieves a cloud based on the parameterization used by Benneke. This parameterization is still a work in progress. Cloud is parameterized by a variable effective radius with a fixed variance of 2; a variable top pressure  $p_{\text{top}}$ ; and a shape factor  $H$  that governs the vertical density of the cloud as follows:  $q(p)=q_0(\log p - \log p_{\text{top}})^H$ . The cloud base is set to a pressure high enough that it isn't seen in transmission. Refractive indices for a particular cloud constituent are required.
- 22) If VARIDENT(IVAR,1) is equal to 333 then the parameter described is a retrieval of the planetary surface gravity parameter:  $\log_{10}(g)$ , where  $g$  is units of  $\text{cm s}^{-2}$ . The next line contains the assumed value of  $\log_{10}(g)$  together with the error.
- 23) If VARIDENT(IVAR,1) is equal to 222 then the Uranus cloud layering scheme of Sromovsky et al. (2011) is used, which specifies 5 distinct cloud layers with 5 optical depths and 3 variable base pressures leading to 8 free parameters. The three lowest cloud decks are very thin and so a specialised atmospheric layering scheme is used to line up the layers with the cloud decks. The 8 following lines contain the variable parameters and the last line contains 5 fixed parameters. Please see `readapriori.f` for more detail.
- 24) If VARIDENT(IVAR,1) is equal to 223 then a revised Uranus cloud layering scheme is used where a depletion of methane in the troposphere is also allowed. The format is the same as for the 222 model, except that there is an extra variable parameter, the methane fraction, and the meaning of two other parameters is slightly changed. Again, please see `readapriori.f` for more detail.
- 25) If VARIDENT(IVAR,1) is equal to 224 then a revised Uranus cloud layering scheme is used where the methane cloud is vertically extendable. The format is the same as for the 223 model, except that there is an extra variable parameter, the methane cloud fractional scale height. Again, please see `readapriori.f` for more detail.
- 26) If VARIDENT(IVAR,1) is equal to 225 then a revised Uranus cloud layering scheme is used where the methane cloud is vertically extendable and a depletion of methane in the troposphere is allowed. The format is the same as for the 224 model, except that there is yet another an extra variable parameter! Again, please see `readapriori.f` for more detail.
- 27) If VARIDENT(IVAR,1) is equal to 227 then the *Crème Brûlée* layering scheme of Sromovsky et al (2017)/Baines et al (2016) is used. Next lines contain the base pressure, opacity and fractional scale height of the main tropospheric cloud layer (TC), then the base pressure and opacity of the chromophore layer, (CB), then the base pressure and opacity of the stratospheric haze layer (SH), all together with errors. Finally, the last line specifies the top pressure of the CB as a fraction of the base pressure (eg. setting this line to 0.2 would make the top pressure of the CB equal to 1/5 of the CB base pressure). To use the default *Crème Brûlée* model, where the top pressure of the CB is equal to 0.9 x the CB base pressure, set this value to 0.
- 28) If VARIDENT(IVAR,1) is equal to 102 then we assume there are two TP profiles (defined by model 29) and we want to compute the weighted average of the spectra calculated using both profiles. Next line contains a priori fraction (0.0 – 1.0) and error. In the retrieval the fraction is limited to remain in this range.

For non-atmospheric parameters (except type 444), then VARIDENT(IVAR,2) has no meaning and is **ignored**. However, if the parameter considered is atmospheric then the third element of VARIDENT(IVAR), i.e., VARIDENT(IVAR,3), is a parameterisation

code for how the profile is to be represented. There are currently over 40 methods and this may be easily extended in future. Currently valid VARIDENT(IVAR,3) codes are:

- 0 Profile is to be treated as continuous over the pressure range of runname.ref, the next line of the .apr file should then contain a filename, which specifies the *a priori* profile as a function of pressure and should have the same number of levels and the same pressures as the .ref file. This filename has the following format:

```

 N CLEN
 P(1) X(1) ERR(1)
 P(2) X(2) ERR(2)
 ...
 P(N) X(N) ERR(N)

```

N must be the same as NPRO defined in runname.ref, and the pressure grid should also be identical. X(1:N) is the *a priori* profile, and ERR(1:N) the associated errors. CLEN contains the assumed correlation length of the profile (in terms of log(P)).

- 1 Profile is to be represented as a deep value up to a certain ‘knee’ pressure, and then a defined fractional scale height. The next line of the .apr file then contains the ‘knee’ pressure, followed by the *a priori* deep and fractional scale height values together with their estimated errors.
- 2 Profile is to be represented by a simple scaling of the corresponding profile runname.ref (for T, v.m.r.), aerosol.ref (for aerosol density), parah2.ref (for para-H<sub>2</sub> fraction) or fcloud.ref (for fractional cloud cover). The next line of the .apr file then contains the *a priori* factor and error.
- 3 Profile is again to be represented by a simple scaling of the corresponding profile runname.ref (for T, v.m.r.), aerosol.ref (for aerosol density), parah2.ref (for para-H<sub>2</sub> fraction) or fcloud.ref (for fractional cloud cover). However, in this option the *a priori* factor and error, contained in the next line of the .apr file, are first converted to log value and fractional error. This ensures that the profile can never go negative since no matter how small the log-value gets, its exponent will still be positive. At the end of the retrieval, the exponent of final log value and error are output to the .mre file.
- 4 Very similar to case when VARIDENT(IVAR,3) = 1 in that the profile is to be represented as a deep value up to a certain ‘knee’ pressure, and then a defined fractional scale height. However, in this case the knee pressure is also a variable parameter and thus must be supplied with an error estimate.
- 5 No longer supported.
- 6 Profile is a cloud profile represented by a base height, optical depth and cloud scale height. The next line of the .apr file then contains the reference altitude, followed by the *a priori* optical depth and scale height values together with their estimated errors. All quantities are taken as logs so negative fractional scale heights are not allowed.
- 7 Very similar to case when VARIDENT(IVAR,3) = 1 in that the profile is to be represented by value at a certain ‘reference’ pressure, and then a defined fractional scale height. However, in this case the profile is extended both upwards and below the reference pressure.
- 8 Profile is a cloud profile represented by a variable base pressure, opacity and fractional scale height. The next line of the .apr file then contains the *a priori*

base pressure, followed by the *a priori* opacity (or integrated particles/cm<sup>2</sup> if the .xsc file has not been normalised at a reference wavelength) and fractional scale height values together with their estimated errors. All quantities are taken as logs so negative fractional scale heights are not allowed.

- 9 Profile is a cloud profile represented by a variable base height, opacity and fractional scale height. The next line of the .apr file then contains the *a priori* base altitude, followed by the *a priori* opacity (or integrated particles/cm<sup>2</sup> if the .xsc file has not been normalised at a reference wavelength) and fractional scale height values together with their estimated errors. Whilst the base height is read linearly, the opacity and fractional scale height are held as log values so negative values for these parameters are not allowed.
- 10 Profile is a condensing cloud. The parameterisation variables contain the deep gas vmr, the required relative humidity above the condensation level, the required opacity of the condensed cloud and the fractional scale height of the condensed cloud. The resulting cloud density will condense in cloud profile defined by VARPARAM(IVAR,1).
- 11 Condensing gas, but no associated cloud. Model requires the deep gas abundance and the desired relative humidity above the condensation level only (VARPARAM(IVAR,1)=1) or at all levels (VARPARAM(IVAR,1)=0).
- 12 Profile is a cloud with a specific density profile that has the shape of a Gaussian line. The profile is parameterised with a peak specific density, the pressure level of that peak and the width of distribution in units of log(pressure). The next line of the .apr file then contains the *a priori* peak specific density (i.e., particles/gram) and error, followed by the *a priori* pressure at the peak and the log-pressure width, with their respective errors. All quantities are taken as logs.
- 13 Profile is a cloud with a specific density profile that has the shape of a Lorentzian line. The profile is parameterised with a peak specific density, the pressure level of that peak and the width of distribution in units of log(pressure). The next line of the .apr file then contains the *a priori* peak specific density (i.e., particles/gram) and error, followed by the *a priori* pressure at the peak and the log-pressure width, with their respective errors. All quantities are taken as logs.
- 14 Profile is a cloud with a specific density profile that has the shape of a Gaussian line. The profile is parameterised with a total opacity, the altitude where the distribution peaks and the width of distribution in units of km. The next line of the .apr file then contains the *a priori* opacity and error, followed by the *a priori* altitude where the distribution peaks and the log width in km, with their respective errors. All quantities are taken as logs, except the altitude of the peak.
- 15 Profile is a cloud with a specific density profile that has the shape of a Lorentzian line. The profile is parameterised with total opacity, the altitude where the distribution peaks and the width of distribution in units of km. The next line of the .apr file then contains the *a priori* opacity and error, followed by the *a priori* altitude where the distribution peaks and the log width in km, with their respective errors. All quantities are taken as logs, except the altitude of the peak.
- 16 Profile is specified by a value at a reference pressure together with a lapse rate (assumed positive and in units of K/km) above and below. This is of most use for temperature. It is assumed that the temperature increases above and below the reference pressure level and so the lapse rate is stored as a positive number in both instances in order that it can be held as a log number. The next line of the .apr file then contains the *a priori* nominal temperature and error at the

reference tropopause pressure, followed on the next line by the *a priori* reference tropopause pressure and error. The next line gives the tropospheric lapse rate and error (i.e., the lapse rate at pressures greater than the reference pressure) while the final line gives the stratospheric lapse rate.

- 17 Very similar to case when VARIDENT(IVAR,3) = 7 in that the profile is represented by value at a certain 'reference' pressure, and then a defined fractional scale height, which extends both upwards and below the reference pressure. However, in this model the profile is limited to not increase or decrease at pressures less than a second specified pressure level (assumed to be less than the reference pressure).
- 18 Very similar to case when VARIDENT(IVAR,3) = 7 in that the profile is represented by value at a certain 'reference' pressure, and a defined fractional scale height. However, in this model the profile is fixed at altitudes above the reference pressure level.
- 19 Very similar to case when VARIDENT(IVAR,3) = 9, where a cloud profile is represented by a variable base height, optical depth and fractional scale height. However, here there is an extra parameter, which forces the profile to zero for pressures less than 0.1 atm (to simulate a tropopause 'lid'). The next line of the .apr file then contains the *a priori* base altitude, followed by the *a priori* abundance and fractional scale height values together with their estimated errors. The final line provides the cut-off parameter and error. All quantities are taken as logs so negative fractional scale heights are not allowed. The extra parameter,  $\alpha$ , is used to provide an additional multiplicative factor to the density:  $x(p) = 1.0 - \exp\left((\log(p) - \log(0.1))/\alpha\right)$ . NB  $x(p) = 0$  for  $p < 0.1$  atm.
- 20 Very similar to case 1 in that profile is to be represented as a deep value up to a certain 'knee' pressure, and then a defined fractional scale height. However, in this parameterisation, the profile is forced to a very small number at pressures less than a 'tropopause' temperature. The next line of the .apr file then contains the 'knee' and 'tropopause' pressures, followed by the *a priori* deep and fractional scale height values together with their estimated errors.
- 21 Similar to model 9. Profile is represented by a value at a variable height, rather than PRESSURE plus a fractional scale height. Below the reference height the profile is set to zero. In addition, this model scales the profile to give the requested integrated cloud optical depth. The radius is set to that assumed/fitted for the associated 444 model input. The fractional scale height is scaled according to the radius (so small particles have larger FSH).
- 22 Temperature profile after Robinson and Catling (2012). Input parameters:
  - a. tau0    optical depth at bottom of atmosphere
  - b. n        tau depends on pressure as  $\tau = \tau_0 * (p/p_0)^n$
  - c. Teff    Radiative equilibrium temperature (sets stratospheric temperature)
  - d. alpha   Modification to DALR to account for SALR. In range 0.7-0.9
  - e. T0       Temperature at bottom of atmosphere

Temperature at top of atmosphere set to Eddington gray approximation, as this extends to lower altitudes, the lapse rate is either limited to the DALR modified by alpha or limited to lapse rate extending up for temperature T0 at P0.
- 23 Profile is defined by two (p,v) points, with a linear gradient (in log p) in between. The low pressure point is at (p1,v1) and the high pressure point is at

(p2,v2). Profile is constant above/below this gradient region (i.e.,  $p < p1$   $v=v1$  and  $p > p2$   $v=v2$ .) All variables are retrieved. Input format:

```
IDGAS IDISO 23
VMR1 VMR1_ERR
P1 P1_ERR
VMR2 VMR2_ERR
P2 P2_ERR
```

- 24 Profile is represented by two separate values above and below a knee pressure (Sromovsky et al, 2017). Next lines of the .apr file respectively contain the profile value below the knee pressure, the profile value above the knee pressure, and the value of the knee pressure itself, together with estimated errors
- 25 Profile is to be treated as continuous as when  $\text{VARIDENT}(\text{IVAR},3) = 0$ , but represented by fewer points than in <runname.ref> to achieve implicit smoothing and faster retrieval times. The next line of the .apr file should then contain a filename, which specifies the *a priori* profile as a function of height using the following format:

```
NP CLN NLayer
P(1) X(1) ERR(1)
P(2) X(2) ERR(2)
...
P(NP) X(NP) ERR(NP)
```

Where NLayer is the number of homogeneous layers to split the profile into. This must be equal to the value of NLayer specified in <runname.set>. If more than one  $\text{VARIDENT}(\text{IVAR},3) = 25$  profile present, the homogeneous layers will be split either according to a separate pressure grid specified in the file 'pressure.lay', or according to the pressure grid found in the first profile listed in <runname.apr>. Unlike in the  $\text{VARIDENT}(\text{IVAR},3) = 0$  case, NP does not need to be equal to NPRO (though both the maximum- and minimum pressure values from <runname.ref> do need to be present to prevent infinity errors).

- 26 Profile is defined by two (p,v) points, with a linear gradient (in log p) in between. The low pressure point is at (p1,v1) and the high pressure point is at (p2,v2). Profile is constant above this gradient region and zero below (i.e.,  $p < p1$   $v=v1$  and  $p > p2$   $v=0$ .) All variables are retrieved. Same input format as 23.
- 27 Step profile. Three lines

```
KNEE_PRESSURE ERR
DEEP_VALUE ERR
SHALLOW_VALUE ERR
```

Can be used for vmr or temperature. If not temperature, then deep/shallow values are handled as logs. Uses a tanh approximation to the step function in a similar way to 24, except pressure is logged to give more consistent step transition (in log(p) ).

- 28 Profile is kept as specified in runname.ref/aerosol.ref, but at one level, where the profile is allowed to vary. The next two lines must contain:

```
LEVEL
X(LEVEL) ERR(LEVEL)
```

Where LEVEL is then index of the point that is allowed to vary (from 1 to NPRO)

- 29 Profile is defined at multiple latitudes and longitudes to enable disc integration over an inhomogeneous disc. Very similar to model 0, but at multiple locations. The filename that this model points to has the following format:

```

NLOCATION N CLEN
 LAT(1) LON(1)
 LAT(2) LON(2)
 ...
 LAT(NLOCATION) LON(NLOCATION)
 P(1) X(1) ERR(1)
 P(2) X(2) ERR(2)
 ...
 P(N) X(N) ERR(N)
 ...
 Same blocks of P,X,ERR for remaining NLOCATION-1 profiles.

```

N must be the same as NPRO defined in runname.ref, and the pressure grid should also be identical. X(1:N) is the *a priori* profile, and ERR(1:N) the associated errors. CLEN contains the assumed correlation length of the profile (in terms of log(P)).

- 30 Profile is defined at multiple longitudes and assumed to tend to a constant mean value at north and south pole. The filename that this model points to has the following format:

```

NLONG N CLEN1 CLEN2
 P(1) X(1,1), ..., (1,NLONG), ERR(1)
 P(2) X(2,1), ..., (2,NLONG), ERR(2)
 P(3) X(3,1), ..., (3,NLONG), ERR(3)
 ...
 P(N) X(N,1), ..., X(N, NLONG), ERR(N)
XPC

```

Where NLONG is the number of longitudes that profile is held for, covering the longitudes 0 to  $NLONG \cdot 360 / (NLONG + 1)$  and N is the number of vertical levels. CLEN1 is the vertical correlation length (in units of log(pressure)), while CLEN2 is the longitudinal correlation length (in degrees). XPC is the coefficient of  $\cos(\text{latitude})$ , i.e.,  $T(\phi) = T(\phi=0) (\cos \phi)^{XPC}$ .

If the first line of this file contains 'Ex', then different errors are assumed as a function of longitudes and the format of the rest of the file is:

```

NLONG N CLEN1 CLEN2
 P(1) X(1,1), ..., (1,NLONG), ERR(1,1), ..., ERR(1,NLONG)
 P(2) X(2,1), ..., (2,NLONG), ERR(2,1), ..., ERR(2,NLONG)
 P(3) X(3,1), ..., (3,NLONG), ERR(3,1), ..., ERR(3,NLONG)
 ...
 P(N) X(N,1), ..., X(N, NLONG), ERR(N,1),..., ERR(N,NLONG)
XPC

```

- 31 Profile is defined to be a multiple of the reference profile, but tends to a constant value at north and south pole. Very similar to model 30. The filename that this model points to has the following format:

```
NLONG CLEN2
X(1), X(2), X(3), ..., X(NLONG), ERR
XPC
```

Where NLONG is the number of longitudes that profile is held for, covering the longitudes 0 to  $\text{NLONG} \times 360 / (\text{NLONG} + 1)$ . CLEN2 is the longitudinal correlation length (in degrees). Here X(I) is the scaling factor (held as logs) for longitude I. XPC is the coefficient of  $\cos(\text{latitude})$ , i.e.,  $T(\phi) = T(\phi=0) (\cos \phi)^{\text{XPC}}$ .

- 32 Similar to model 8 in that profile is a cloud profile represented by a variable base pressure, specific density at the level and fractional scale height. The next line of the .apr file then contains the *a priori* base pressure, followed by the *a priori* opacity and fractional scale height values together with their estimated errors. All quantities are taken as logs so negative fractional scale heights are not allowed. Difference from Model 8 is that cloud density at pressures greater than the base pressure is set to drop exponentially with increasing pressure with a scale height of 1km, rather than just being set to zero. This makes it easier for NEMESIS to actually find an optimal value of the knee pressure.
- 33 Profile is defined at multiple longitudes and assumed to tend to a constant mean value at north and south pole. The filename that this model points to has the following format:

```
NLONG N CLEN1 CLEN2
P(1) X(1,1), X(2,1), X(3,1), ..., X(N,1), ERR(1)
P(2) X(1,2), X(2,2), X(3,2), ..., X(N,2), ERR(1)
P(3) X(1,3), X(2,3), X(3,3), ..., X(N,3), ERR(1)
...
P(N) X(1,NLONG), X(2, NLONG), X(3, NLONG), ..., X(N, NLONG),
ERR(NLONG)
FRAC(1), FRAC(2), ..., FRAC(NLONG), EFRAC
```

Where NLONG is the number of longitudes that profile is held for, covering the longitudes 0 to  $\text{NLONG} \times 360 / (\text{NLONG} + 1)$ , and N is the number of vertical levels. CLEN1 is the vertical correlation length (in units of  $\log(\text{pressure})$ ), while CLEN2 is the longitudinal correlation length (in degrees). FRAC is the required fraction,  $f$ , used to compute the parameter as a function of latitude as:  $T(\phi) = T(\phi=0) (f(\cos \phi)^{0.25} + (1-f)(\cos \phi)^2)$ .

- 34 Profile is the Milne-Eddington temperature profile. The next line of the .apr file then contains the *a priori* bolometric temperature (plus error), followed by the *a priori* pressure scaling factor (plus error).

- 35 Very similar to model 30, except that the profile is defined at multiple **latitudes** and assumed to be invariant with longitude. The filename that this model points to has the following format:

```
NLAT N CLEN1 CLEN2
P(1) X(1,1), X(2,1), X(3,1), ..., X(N,1), ERR(1)
P(2) X(1,2), X(2,2), X(3,2), ..., X(N,2), ERR(1)
P(3) X(1,3), X(2,3), X(3,3), ..., X(N,3), ERR(1)
...
P(N) X(1,NLAT), X(2, NLAT), X(3, NLAT), ..., X(N, NLAT), ERR(NLAT)
```

Where NLAT is the number of latitudes that profile is held for, covering the latitudes -90 to 90 and N is the number of vertical levels. CLEN1 is the vertical correlation length (in units of log(pressure)), while CLEN2 is the latitudinal correlation length (in degrees).

- 36 Very similar to model 31 and 30, except that the profile is defined to be a multiple of the reference profile for a range of **latitudes**. The filename that this model points to has the following format:

```
NLAT CLEN2
X(1), X(2), X(3), ..., X(NLAT), ERR
```

Where NLAT is the number of latitudes that profile is held for, covering the latitudes -90 to 90. CLEN2 is the latitudinal correlation length (in degrees).

- 37 Cloud which has constant opacity/bar between two specified pressure levels (measured in bar). The next line of the .apr file then contains the two pressures (in bar) in the order high - low, followed by the a priori opacity/bar and error.
- 38 Karkoschka and Tomasko (2011) Neptune methane profile. Scaling parameter of methane abundance profile at depth. User enters required deep CH<sub>4</sub> slope ( $100 \cdot d\_ch4\_vmr/dpress$ ) and error. In the following line the user also specifies the required (and fixed) values of the tropospheric methane VMT, the stratospheric methane VMR and intermediate relative humidity (RH: 0 - 1).
- 39 Irwin et al. (2020) Neptune methane profile. User enters required deep CH<sub>4</sub> vmr and error. In the following line the user also specifies the required (and fixed) values of the stratospheric methane VMR and intermediate relative humidity (RH: 0 - 1).
- 40 Model to simulate haze photochemical models of Toledo et al. User enters the base aerosol specific density (plus error) and fractional scale height (plus error). Next line contains p1, p2 and p3, which are held in the VARPARAM array. Between pressures p1 and p2, the specific density of cloud type ABS(VARIDENT(IVAR,1)) is set to match the base density and fractional scale height. Between p2 and p3 this specific density profile is associated with cloud type ABS(VARIDENT(IVAR,1))+1. Basically, this ties the specific density profiles of the two cloud types together, but allows particles in the upper layer

to have different scattering properties from those in the lower layer. Outside  $p_1$  to  $p_3$ , the specific densities of both cloud types are set to  $1E-36$ . User enters:

Deep cloud density and error  
 Cloud fractional scale height and error  
 $p_1, p_2, p_3$

- 41 Descended methane profile of Sromovsky et al. (2019) for ice giant planets.

User enters:

Deep methane vmr and error  
 Deep pressure (bar) and error  
 Cloud-top relative humidity and error  
 Tropopause relative humidity and error  
 Descended VX parameter and error  
 Stratospheric limiting VMR and pressure of tropopause (bar)

- 42 Ackerman and Marley vmr and cloud model. User enters:

Deep vmr and error;  
 Upwards heat flux ( $W\ m^{-2}$ ) and error;  
 $f_{rain}$  parameter and error.

This is followed by 6 parameters in one row:

- a) Imodel (0=Lewis, 1=Ackerman and Marley);
- b) JCONT (associated cloud ID);
- c) DENScond (density of condensate in  $g/cm^3$ );
- d) RADCOND (radius of condensate in microns);
- e) MWCOND (molecular weight of condensate in g);
- f) XCORR (reference extinction x-section of particles ( $cm^2$ )).

The last parameter is necessary as in most cases the .xsc files are normalised to a reference wavelengths.

- 43 Double-grey analytic T-P profile of Line et al. (2013) and, later, Parmentier and Guillot (2014). User enters:

$\alpha$  parameter and error  
 $\beta$  parameter and error  
 $\kappa_{IR}$  parameter and error  
 $\gamma_1$  parameter and error  
 $\gamma_2$  parameter and error

Followed by 4 parameters in one row:  $T_{star}$  (K),  $R_{star}$  (km),  $S_{dist}$  (km) and  $T_{int}$  (K). Could still update  $R_{star}$  and  $S_{dist}$  to be read from .sol and .set files, respectively.

- 44 Profile is defined at multiple longitudes and assumed to tend to a constant mean value at north and south pole. Model uses double-grey analytic T-P profile of Line et al. (2013) and, later, Parmentier and Guillot (2014). The filename that this model points to has the following format:

```
NLONG CLen2
ALPHA(1), ..., ALPHA(NLONG), ERR_ALPHA
BETA(1), ..., BETA(NLONG), ERR_BETA
```

K(1), ..., K(LONG), ERR\_KAPPA  
 GAMMA1(1), ..., GAMMA1(NLONG), ERR\_G1  
 GAMMA1(2), ..., GAMMA2(NLONG), ERR\_G2  
 XPC,  
 TSTAR, RSTAR, SDIST, TINT

Where NLONG is the number of longitudes that profile is held for, covering the longitudes 0 to  $\text{NLONG} \times 360 / (\text{NLONG} + 1)$ . CLLEN2 is the longitudinal correlation length (in degrees). XPC is the coefficient of  $\cos(\text{latitude})$ , i.e.,  $T(\phi) = T(\phi=0) (\cos \phi)^{\text{XPC}}$ . TSTAR, RSTAR, SDIST and TINT are the remaining parameters for this model.

If the first line of this file contains 'Ex', then different errors are assumed as a function of longitudes and the format of the rest of the file is:

NLONG          CLLEN2  
 ALPHA(1), ..., ALPHA(NLONG), ERR\_A(1), ..., ERR\_A(NLONG)  
 BETA(1), ..., BETA(NLONG), ERR\_B(1), ..., ERR\_B(NLONG)  
 K(1), ..., K(LONG), ERR\_K(1), ..., ERR\_K(NLONG)  
 GAMMA1(1), ..., GAMMA1(NLONG), ERR\_G1(1), ..., ERR\_G1(NLONG)  
 GAMMA1(2), ..., GAMMA2(NLONG), ERR\_G2(1), ..., ERR\_G2(NLONG)  
 XPC,  
 TSTAR, RSTAR, SDIST, TINT

- 45 As Model 39 in that this is the Irwin et al. (2020) Neptune methane profile. However, in this case all parameters are variable. User enters required deep CH<sub>4</sub> vmr and error, followed by relative humidity (RH: 0 - 1) and error, and then stratospheric methane VMR and error.
- 46 As model 14, but for a double peaked structure. Each part of the profile is parameterised with a total opacity, the altitude where the distribution peaks and the width of distribution in units of km. The next line of the .apr file then contains the *a priori* opacity and error, followed by the *a priori* altitude where the distribution peaks and the log width in km, with their respective errors. All quantities are taken as logs, except the altitude of the peak.
- 47 As model 14, but for a cloud centred at a specified pressure (rather than altitude), variable FWHM (log pressure units) and defined total opacity. The next line of the .apr file then contains the *a priori* opacity, the *a priori* pressure where the distribution peaks, and the *a priori* width (in units of log pressure), with their respective errors. All quantities are taken as logs.
- 48 As model 32, in that profile is a cloud profile represented by a variable base pressure, specific density at the level, fractional scale height, but also a variable top pressure. The next line of the .apr file then contains the *a priori* base pressure, followed by the *a priori* top pressure, opacity and fractional scale height values together with their estimated errors.

- 49 Profile is a multiplier of another gas or cloud profile. Useful if, for example, you want to set the CH<sub>3</sub>D profile as a multiple of the CH<sub>4</sub> profile, which is a variable and varies with height. Or if you want a cloud profile that is a multiple of another. First line contains the ID, ISO of the profile you want to multiply. If  $ID < 0$  then it is assumed that you want to multiply the cloud profile:  $\text{mod}(ID)$ . The second line contains the multiplying factor and error.
- 50 As model 13, but for a cloud centred at a specified pressure (rather than altitude), variable FWHM (log pressure units) and defined total opacity. The next line of the .apr file then contains the *a priori* opacity, the *a priori* pressure where the distribution peaks, and the *a priori* width (in units of log pressure), with their respective errors. All quantities are taken as logs.
- 51 Similar to model 47, but here a cloud is specified by its integrated opacity, the cloud top pressure, the pressure difference between the top and bottom pressures (log pressure units) and the FWHM (also log pressure units) at which the opacity drops off above the top and below the bottom of the cloud. The specific density of the cloud is constant between the top and bottom pressures. The next lines of the .apr file then contains the *a priori* opacity, the *a priori* cloud top pressure, the *a priori* difference between the top and bottom cloud pressures (log pressure) and the *a priori* width (log pressure) of the gaussian fall off above and below the cloud, all with their respective errors. All quantities are taken as logs.
- 52 Similar to model 47, with cloud centred at a specified pressure, variable FWHM (log pressure units) and defined total opacity, but here the cloud has a different, variable FWHM below the specified pressure, and also a cutout to prevent this cloud significantly overlapping with an upper haze layer. The next lines of the .apr file then contain: 1) the *a priori* opacity; 2) the *a priori* pressure where the distribution peaks; 3) the *a priori* width above the cloud peak (in units of log pressure); and 4) the *a priori* width below the cloud peak (in units of log pressure), all with their respective errors. Finally, the fixed peak pressure and FWHM of the haze above this cloud deck is given. The opacity of the cloud is forced to zero where it overlaps the upper cloud and does not increase again at higher altitudes. All quantities are taken as logs.
- 53 Very similar to case when  $\text{VARIDENT}(\text{IVAR},3) = 7$  in that the profile is to be represented by value at a certain 'reference' pressure, and then a defined fractional scale height both above and below this level. However, the deep value is also limited to not exceed a user-defined limit (and error).
- 54 Similar to model 52, with cloud centred at a specified pressure, variable FWHM (log pressure units) and defined total opacity, but here the cloud has a different, variable FWHM below the specified pressure, and the density falls off as an exponential above the base pressure, rather than a gaussian, to better mimic the main cloud seen by the Galileo probe nephelometer. The next lines of the .apr file then contain: 1) the *a priori* opacity; 2) the *a priori* pressure where the distribution peaks; 3) the *a priori* width above the cloud peak (in units of log pressure); and 4) the *a priori* width below the cloud peak (in units of log pressure), all with their respective errors. All quantities are taken as logs.

- 55 Similar to model 17, in that the profile is represented by value at a certain 'reference' pressure, and then a defined fractional scale height, which extends both upwards and below the reference pressure. The vmr is limited to be within variable upper and lower limits. The next lines of the .apr file then contain: 1) the reference pressure; 2) the *a priori* abundance at that pressure; 3) the *a priori* fractional scale height; 4) the limiting lower *a priori* abundance; and 5) the limiting upper *a priori* abundance. All quantities are taken as logs.
- 56 Similar to model 11 for a condensing gas, and no associated cloud. Model requires the deep gas abundance, middle gas abundance (above a fixed pressure level) and the desired maximum relative humidity. The next lines of the .apr file then contain: 1) the *a priori* deep abundance and error; 1) the *a priori* middle abundance and error; 3) the *a priori* relative humidity (RH) and error; and 4) the reference pressure  $p_{\text{knee}}$  and a condensation flag ICOND. If ICOND=0 then the gas partial pressure cannot exceed  $\text{RH} \cdot P_{\text{SVP}}$  at any level. If ICOND=1 then the gas partial pressure cannot exceed  $P_{\text{SVP}}$ , but once above the condensation level the gas partial pressure is set to  $\text{RH} \cdot P_{\text{SVP}}$ . This latter model is similar to the parameterisation used by Brooke et al. (1998).
- 57 Gas profile where the abundance is constant up to a knee pressure and then falls linearly with respect to log pressure to a second abundance defined at a lower stratospheric pressure. The profile is further linearly extrapolated to lower pressures. First line defines the pressures PKNEE, PSTRAT. Second line gives *a priori* deep abundance and error. Third line gives stratospheric abundance and error.
- 58 Profile is represented by a value at a variable pressure level plus a fractional scale height. Below the knee pressure the profile is set to drop exponentially. Thickness is total opacity, not the specific density at  $p_{\text{knee}}$ . Like model 32, but parameter order is swapped. The next lines contain: 1) the *a priori* opacity; 2) the *a priori* pressure where the distribution peaks; 3) the *a priori* fractional scale height above cloud base.
- 59 Condensing gas profile, which has a deep abundance, middle abundance above knee pressure, fractional scale height in middle region and maximum relative humidity. The next lines contain: 1) the *a priori* deep abundance; 2) the *a priori* middle abundance; 3) the *a priori* fractional scale height above knee pressure; 4) the *a priori* relative humidity; and 5) the knee pressure (fixed).
- 60 Like model 51, but cloud is specified by its maximum specific density, the cloud top pressure, the pressure difference between the top and bottom pressures (log pressure units) and the FWHM (also log pressure units) at which the opacity drops off above the top and below the bottom of the cloud. The specific density of the cloud is constant between the top and bottom pressures. The next lines of the .apr file then contains the *a priori* specific density, the *a priori* cloud top pressure, the *a priori* difference between the top and bottom cloud pressures (log pressure) and the *a priori* width (log pressure) of the gaussian fall off above and below the cloud, all with their respective errors. All quantities are taken as logs.

- 61 Like model 60, but cloud is specified by its maximum specific density at an upper pressure level, the fsh below that, a bottom cloud pressure and then width of the gaussian cut-off above and below. The next lines of the .apr file then contains the *a priori* specific density, the *a priori* cloud top pressure, the *a priori* cloud bottom pressure, the *a priori* fractional scale height between these levels, and finally, the gaussian fall off above and below the cloud, all with their respective errors. All quantities are taken as logs.
- 62 Like model 56. Model requires the deep gas abundance, middle gas abundance (above a fixed pressure level), desired maximum relative humidity after condensation, and then rate of fall-off of RH above that level. The next lines of the .apr file then contain: 1) the *a priori* deep abundance and error; 2) the *a priori* middle abundance and error; 3) the *a priori* relative humidity (RH) and error; 4) the scale height of the RH above the condensation level; and 5) the fixed pressure level.
- 63 Like model 62, but also condenses a cloud. Model requires the deep gas abundance, middle gas abundance (above a fixed pressure level), desired maximum relative humidity after condensation, and then rate of fall-off of RH above that level. Then we have the opacity of the associated cloud and the fsh of that cloud above the condensation level, and various fixed parameters. The next lines of the .apr file contain: 1) the *a priori* deep abundance and error; 2) the *a priori* middle abundance and error; 3) the *a priori* relative humidity (RH) and error; 4) the scale height of the RH above the condensation level and error; 5) the cloud opacity and error; 6) the cloud scale height and error. Then we have PKNEE and ICONT (i.e., pressure of deep step, and ID of condensing cloud), and finally we have PHAZE and WHAZE which reduces the cloud opacity to zero at levels above PHAZE, with a gaussian cut-off width of WHAZE.
- 64 Like model 63, but the gas abundance between PKNEE and PCOND can fall with height. The next lines of the .apr file contain: 1) the *a priori* deep abundance and error; 2) the *a priori* middle abundance and error; 3) the fractional scale height of above PKNEE and error; 4) the *a priori* relative humidity (RH) and error; 5) the scale height of the RH above the condensation level and error; 6) the cloud opacity and error; 7) the cloud scale height and error. Then we have PKNEE and ICONT (i.e., pressure of deep step, and ID of condensing cloud), and finally we have PHAZE and WHAZE which reduces the cloud opacity to zero at levels above PHAZE, with a gaussian cut-off width of WHAZE.
- 65 Like model 64, but model includes a variable offset between the condensation level and the condensed cloud, to match the ‘C-shaped’ cloud profiles familiar from terrestrial studies. The next lines of the .apr file contain: 1) the *a priori* deep abundance and error; 2) the *a priori* middle abundance and error; 3) the fractional scale height above PKNEE and error; 4) the *a priori* relative humidity (RH) and error; 5) the scale height of the RH above the condensation level and error; 6) the cloud opacity and error; 7) the cloud scale height and error; 8) the offset of the cloud base from the condensation level. Then we have PKNEE and ICONT (i.e., pressure of deep step, and ID of condensing cloud), and finally we

have PHAZE and WHAZE which reduces the cloud opacity to zero at levels above PHAZE, with a gaussian cut-off width of WHAZE.

- 66 Like model 64, but this model has two associated cloud profiles, one at a variable PCLOUD, where the gas vmr first drops, and then at the condensation level. The next lines of the .apr file contain: 1) the *a priori* gas deep abundance and error; 2) the *a priori* middle gas abundance and error at pressures less than PCLOUD; 3) the fractional scale height of above PCLOUD and error; 4) the *a priori* relative humidity (RH) and error; 5) the scale height of the RH above the condensation level and error; 6) the 1<sup>st</sup> cloud opacity and error; 7) the 1<sup>st</sup> cloud scale height and error; 8) the 1<sup>st</sup> cloud base pressure (PCLOUD) and error; 9) the 2nd cloud opacity and error; 10) the 2<sup>nd</sup> cloud scale height and error; Then we have PHAZE1 and WHAZE1 which reduces the 1<sup>st</sup> cloud opacity to zero at levels above PHAZE1, with a gaussian cut-off width of WHAZE1. Then we have PHAZE2 and WHAZE2 for the second cloud.

Further parameterisation schemes may be defined in the future as required. Any additional parameters (e.g., the knee pressure for VARIDENT(IVAR,3) = 1 to 4) are held in the VARPARAM(NVAR,NPARAM) array.

Routines that need modifying are:

```
logflag.f
npvar.f
readapriori.f
stripvar.f
subprofretg.f
```

### 3.2 Input .inp file

This file contains specific run information and used to be read in from the standard input. The format is:

---

```
ISPACE, ISCAT, ILBL
WOFF
ENAME
NITER
PHILIMIT
NSPEC, IOFF
LIN
IFORM (optional)
PERCBOOL (optional)
```

---

ISPACE is the wavelength space in which to calculate the spectra and in which the k-tables are tabulated. 0 = wavenumber (cm<sup>-1</sup>) and 1 = wavelength (μm). N.B. all other tabulated spectra files (e.g., '.xsc', '.sur', 'hgphase.dat' etc) should be in the wavespace specified by ISPACE.

ISCAT = 1 indicates whether a multiple scattering calculation is required. If ISCAT = 0, then a thermal emission calculation (with addition of ground radiance for non-giant

planets) is assumed. For scattering calculations the non-gradient forward model is used. (It was originally hoped to update the 'gradient' method into the scattering code, but this turns out not to give any speed advantage). If  $ISCAT = 2$ , then the internal scattered radiation field is calculated first (required for limb-scattering calculations). If  $ISCAT = 3$ , then a single scattering plane-parallel atmosphere calculation is performed. If  $ISCAT = 4$ , then a single scattering spherical atmosphere calculation is performed. If  $ISCAT = 5$ , then an internal flux calculation is performed, in which case the .spx file contains the downwards radiances at several consecutive levels.

$ILBL = 0$  indicates that a correlated-K calculation is required.  $ILBL=1,2$  indicates a line-by-line calculation. **This is an important change from previous versions.** Note that for NemesisPT, this third integer actually sets  $INUMERIC$ , which determines whether the code calculates the functional derivatives using implicit differentiation or numerically.  $ILBL=1$  indicates a line-by-line calculation from scratch, while  $ILBL=2$  indicates a line-by-line calculation using pre-tabulated monochromatic absorption look-up tables, calculated with the program `Calc_lbtable`.

$WOFF$  is any wavenumber/wavelength calibration error, which needs to be added to the synthetic spectra.

$ENAME$  is the name of the file which contains the forward modelling errors to be added to the measurement covariance matrix. The file starts with the number of wavelengths followed by two columns: wavenumber/wavelength and noise. This file is subsequently interpolated to required output wavelengths.

$NITER$  is the number of iterations of the retrieval model required

$PHILIMIT$  is the percentage convergence limit. If the percentage reduction of the cost function  $\Phi$  between iterations is less than  $PHILIMIT$  then the retrieval is deemed to have converged, and the retrieval terminated early.

$NSPEC$  is the total number of retrievals to perform (for measurements contained in the <runname.spx> file.  $IOFF$  is the index of the first spectrum to fit. For example, the <runname.spx> file may contain two sets of observations and you only want to retrieve the second, in which case,  $IOFF = 2$ , and  $NSPEC = 1$ .

$LIN$  is an integer indicating whether the results of a previous retrieval run are to be used to set any of the model atmospheric profiles, and if so how. For example, you might want to retrieve temperature first with one set of wavelengths and subsequently fit gas abundances from another set. Previous retrievals are read in from a '.pre' file (which is direct copy of the '.raw' file of the previous retrieval). The same number ( $IOFF$ ) of retrievals is skipped as in the '.spx' file.

If  $LIN$  is set to 0, then no previous retrievals are read in.

If  $LIN$  is set to 1, then the previous retrievals (by copying <runname>.raw to <runname.pre>) are used both to fix the relevant variables at their last-retrieved value, and also to calculate the effect that their retrieval errors have on the current retrieval by calculating and adding this to the measurement covariance matrix  $S_e$ . Here  $S_x$  is the previously retrieved covariance matrix of

the constituent, and  $\mathbf{K}_n$  is the Jacobian for that constituent calculated for the new wavelength array specified by the current measurement .spx file.

If LIN is set to 2, then the previous retrievals are read in, and if any variable is the same as one of the variables to be retrieved (as listed in the .apr file) then the *a priori* elements and covariance matrix are set to these last retrieved values.

If LIN is set to 3, then the previous retrievals are read in, and if any variable is the same as one of the variables to be retrieved (as listed in the .apr file) then the *a priori* elements and covariance matrix are set to these last retrieved values. In addition, all other parameters are fixed to their last-retrieved value, and their retrieval errors used to modify the measurement covariance matrix  $\mathbf{S}_\epsilon$  as per LIN=1.

IFORM defines the unit of the calculated spectrum and is defined in Section 0.3. The default value is equal to 0.

PERCBOOL defines whether the data in ENAME is given as a systematic offset in radiance values regardless on the brightness of the spectrum, or if it is given as a percentage error on the radiance of the observed spectrum. If PERCBOOL is not specified, the former is assumed, otherwise PERCBOOL must be set to *true*.. Note that if PERCBOOL is explicitly specified in the .inp file, then IFORM cannot be omitted even if it is set to the default value.

### 3.3 Spectrum .spx file

This file contains the spectrum to be fitted together with FOV averaging details. It has a similar format to its .spc predecessor but includes improved FOV averaging locations and weights (which need to be generated off-line).

---

```
FWHM, LATITUDE, LONGITUDE, NGEOM
NCONV(1)
NAV(1)
FLAT(1,I), FLON(1,I), SOL_ANG(1,I), EMISS_ANG(1,I), AZI_ANG(1,I),
 WGEOM(1,I)
...
FLAT(1,NAV(1)), FLON(1,NAV(1)), SOL_ANG(1,NAV(1)),
 EMISS_ANG(1,NAV(1)), AZI_ANG(1,NAV(1)), WGEOM(1,NAV(1))
VCONV(1,1), Y(1,1), ERR(1,1)
VCONV(1,2), Y(1,2), ERR(1,2)
...
VCONV(1,NCONV), Y(1,NCONV), ERR(1,NCONV)
NCONV(2)
NAV(2)
FLAT(2,I), FLON(2,I), SOL_ANG(2,I), EMISS_ANG(2,I), AZI_ANG(2,I),
 WGEOM(2,I)
...
```

```

FLAT(2,NAV(2)), FLON(2,NAV(2)), SOL_ANG(2,NAV(2)),
 EMISS_ANG(2,NAV(2)), AZI_ANG(2,NAV(2)), WGEOM(2,NAV(2))
VCONV(2,1), Y(2,1), ERR(2,1)
VCONV(2,2), Y(2,2), ERR(2,2)
...
VCONV(2,NCONV), Y(2,NCONV), ERR(2,NCONV)
.
.
... weights, angles and spectra repeated for NGEOM spectra in total.

```

---

**FWHM** is the full-width half-maximum of the square box to be convolved with the CIRSADG calculated spectrum. **N.B.** If FWHM is negative, then channel-integration is assumed and a <runname>.fil file must also be present which contains the channel filter functions. The format of this file is straightforward and is best explained by looking at the subroutine `wavesetb.f`, which reads in the filter function and then determines the wavenumbers/wavelengths in the k-tables for which the radiances need to be calculated in order to perform the channel integration. If FWHM is set to zero, then **NEMESIS** assumes that channel-integrated k-tables have been defined and so no further convolution is required.

**LATITUDE** is the planetocentric latitude at the centre of the FOV

**LONGITUDE** is the longitude at the centre of the FOV

**NGEOM** is the number of different observation geometries under which the location is observed.

**NCONV** is the number of convolution wavenumbers/wavelengths in each spectrum. This must at present be the same for all spectra pertaining to a given location on the planet.

**NAV** For each of the NGEOM spectra, NAV defines how many individual spectral calculations need to be performed to construct the field-of-view-averaged spectrum.

For each viewing geometry (NGEOM in total), the parameters NCONV and NAV are first read in. NCONV is the number of convolution wavenumbers/wavelengths in each spectrum, which do not now have to be the same for all NGEOM spectra. NAV specifies how many individual spectra need to be calculated and averaged to simulate the measured field-of-view-averaged spectrum. The next NAV lines contain the integration point latitudes (FLAT), longitudes (FLON), viewing angles (SOL\_ANG, EMISS\_ANG, AZI\_ANG) and weights (WGEOM). The angle definitions are outlined in Figure 1. Following this, there then follows the actual measured wavelengths/wavenumbers, spectrum and errors (all of length NCONV) which are read in and put in total measurement vector **y**. The measurement covariance matrix is here assumed to be diagonal, with variances equal to  $ERR^2$ .

If simulating a Selective Chopper Radiometer (SCR) then for each spectrum, the first NCONV/2 rows list the sideband radiances and the second NCONV/2 rows list the wideband radiances. The wavenumbers/wavelengths of these two blocks should match.

Note that there is a special condition setup for performing integration along the central meridian using the Minnaert limb-darkening approximation. This calculation is performed if NAV=2 and WGEOM(1) and WGEOM(2) are both < 0. In this case

NEMESIS calculates two spectra at the two different viewing geometries, first nadir (viewing and solar),  $I_0$ , and then at one of the scattering quadrature zenith angles (viewing=solar),  $I_1$ . These two spectra can then be used to compute the Minnaert limb-darkening coefficient  $k = \frac{1}{2} \left( 1 + \frac{\log(I_1/I_0)}{\log(\mu)} \right)$ . Given  $I_0$ , and  $k$ , the line-average can be interpolated from a pre-computed table, 'lineminnaert.txt', in /raddata.

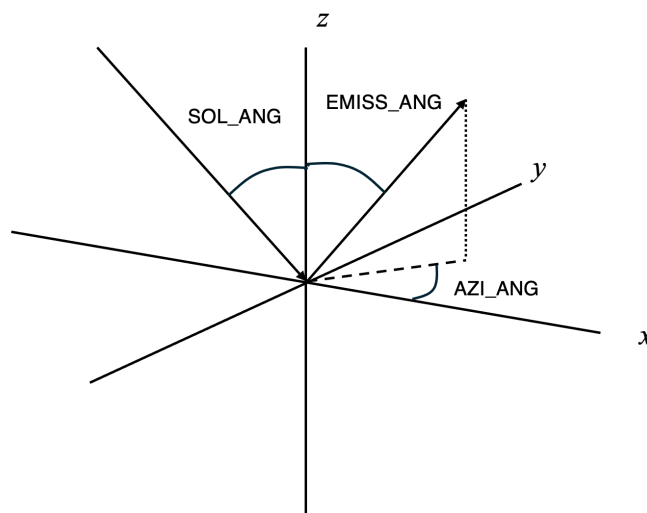


Figure 1. Definition of Viewing Angles used by **NEMESIS** and **RADTRANS**  
When reading in the viewing angles, a negative emission zenith angle (EMISS\_ANG) indicates that the observed spectrum is actually a limb observation where the solar zenith angle (SOL\_ANG) then contains the tangent altitude (km). The definition of these angles is explained in Fig. 1. N.B. Setting AZI\_ANG=0 implies FORWARD scattering.

For ISCAT=5 calculations, SOL\_ANG is as defined, but the EMISS\_ANG variable now contains the height in the atmosphere (km) where the downwards radiance is measured (for Venera Venus flux calculations).

### 3.4 Setup .set file

This file contains scattering quadrature information (if a scattering run is being performed) and layering information. A typical example is:

---

\*\*\*\*\*

```

Number of zenith angles : 5
0.165278957666387 0.327539761183898
0.477924949810444 0.292042683679684
0.738773865105505 0.224889342063117
0.919533908166459 0.133305990851069
1.000000000000000 2.22222222222222D-002
Number of fourier components : 0
Number of azimuth angles for fourier analysis : 100
Sunlight on(1) or off(0) : 1
Distance from Sun (AU) : 5.200
Lower boundary cond. Thermal(0) Lambert(1) : 1
Ground albedo : 0.000
Surface temperature: 150.0

```

```

Alt. at base of bot.layer (not limb) : -40.000
Number of atm layers : 100
Layer type : 1
Layer integration : 1

```

---

The meaning of the fields should be fairly clear. The first half of the file contains setup information for a scattering run, which can now be performed with **NEMESIS**, although the Jacobian has to be calculated numerically. The second half contains information on how the atmosphere is to be split into layers by `subpathg.f` with the layering codes equal to `LAYHT`, `NLAYER`, `LAYTYP` and `LAYINT` respectively and defined in the Radtran manual [R2]. `LAYHT` is used as set in the `.set` file UNLESS a limb-observing geometry is indicated by the `.spx` spectral observation file. In this special case `LAYHT` is set to `SOL_ANG` (section 3.3). Note that if `GALB < 0`, then the albedo is set to 1.0 minus the emissivity defined in the emissivity (`.sur`) file.

### 3.5 Fractional cloud cover file `fcloud.prf` format

If aerosols are defined then **RADTRANS** (and thus **NEMESIS**) needs to know if the cloud is in the form of a uniform thin haze or is instead arranged in thicker clouds covering a certain fraction of the mean area. These details are supplied in the file `fcloud.prf` file.

The first line contains the number of profile levels and the number of cloud particle types (which should match that defined in the `.xsc` file and dust profile file). The following lines then contain the profile heights, fractional cloud cover and identifiers as to which cloud particle types contribute to the fractional cloud. Hence, the first few lines of the `fcloud.prf` file appear as:

```

NPRO, NCONT
HEIGHT(1), FRAC(1), ICLOUD(1,1), ICLOUD(2,1),...,ICLOUD(NCONT,1)
HEIGHT(2), FRAC(2), ICLOUD(1,2), ICLOUD(2,2),...,ICLOUD(NCONT,2)
...
HEIGHT(NPRO), FRAC(NPRO),ICLOUD(1,NPRO),ICLOUD(2,NPRO),...,
ICLOUD(NCONT,NPRO)

```

If `ICLOUD(I,J)` is set to 1, then aerosol type `I` contributes to the broken cloud at level `J`, which has a fractional cloud cover of `FRAC(J)`. If `ICLOUD(I,J)` is set to 0, then aerosol `I` is treated as being part of a uniform haze.

### 3.6 Reference Solar/Stellar Spectrum `.sol` File.

This file (`<runname>.sol`) contains the name of the solar or stellar spectrum file, which is assumed to reside in the `raddata/` directory.

The solar/stellar spectral file format is as follows. The file can contain as many header lines as necessary, each line beginning with a `#` character.

The first line after the header contains the wavenumber/wavelength space of the spectrum: 0 = wavenumber, 1 = wavelength.

The next line contains the radius of the Sun/star in units of km.

The rest of the file contains the wavelength/wavenumbers and spectral luminosity in two columns. Wavelengths/wavenumbers are in units of  $\mu\text{m}$ , or  $\text{cm}^{-1}$ . Spectral luminosity is in units of  $\text{W } \mu\text{m}$ , or  $\text{W } (\text{cm}^{-1})^{-1}$ .

### 3.7 Collision-induced absorption .cia file.

This file (<runname.cia>) contains the name of the CIA file to be used (assumed to exist in the raddata directory of radtrancode).

The first line contains the name of the CIA file. The CIA file is always in wavenumber space.

The second line defines the wavenumber step, dnu, of the CIA table.

The third line gives the number of para- $\text{H}_2$  fractions listed, NPARA.

CIA tables are in two formats, one which lists  $\text{H}_2\text{-H}_2$  (eqm),  $\text{H}_2\text{-He}$ (eqm),  $\text{H}_2\text{-H}_2$  (normal),  $\text{H}_2\text{-He}$  (normal) and then 5 other pairs:  $\text{H}_2\text{-N}_2$ ,  $\text{H}_2\text{-CH}_4$ ,  $\text{N}_2\text{-N}_2$ ,  $\text{CH}_4\text{-CH}_4$  and  $\text{H}_2\text{-CH}_4$ . The other type of CIA table lists  $\text{H}_2\text{-H}_2$  and  $\text{H}_2\text{-He}$  only, but for a number of different ortho/para fractions. For the usual CIA table format, NPARA should be set to zero. For a variable para- $\text{H}_2$  CIA table, NPARA can be set to be between 0 and 24. The exact number depends on the table itself.

### 3.8 Additional flags .fla file.

This file (<runname.fl>) contains the following integer flags that used to be hardwired in different parts of the code:

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INORMAL | 0 or 1 depending on whether the ortho/para- $\text{H}_2$ ratio is in equilibrium (0) or normal 3:1 (1).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| IRAY    | Sets the Rayleigh optical depth calculation:<br>0 = Rayleigh scattering optical depth is not included.<br>1 = Rayleigh optical depths suitable for gas giant atmosphere<br>2 = Rayleigh optical depths suitable for $\text{CO}_2$ dominated atmosphere<br>3 = Rayleigh optical depths suitable for a $\text{N}_2\text{-O}_2$ atmosphere.<br>4 = New Rayleigh optical depths suitable for a gas giant atmosphere.<br>5 = New Rayleigh optical depths suitable for a gas giant atmosphere PLUS Raman scattering.<br>6 = New Rayleigh optical depths suitable for a gas giant atmosphere PLUS Sromovsky Raman scattering, PLUS Sromovsky Polarisation Correction. |
| IH2O    | Turns additional $\text{H}_2\text{O}$ continuum off (0) or on (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| ICH4    | Turns additional $\text{CH}_4$ continuum off (0) or on (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| IO3     | Turns additional $\text{O}_3$ UV continuum off (0) or on (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| INH3    | Turns additional $\text{NH}_3$ continuum off (0) or on (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| IPTF    | Used in only a few routines to switch between normal partition function calculation (0) or the high-temperature partition function for $\text{CH}_4$ for Hot Jupiters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| IMIE    | Only relevant for scattering calculations. If set to 0, the phase function is computed from the associated Henyey-Greenstein hgphase*.dat files. However, if set to 1, the phase function is computed from the Mie-Theory calculated PHASEN.DAT file.                                                                                                                                                                                                                                                                                                                                                                                                          |
| IUVSCAT | Used for ultraviolet reflectance spectra, otherwise set to (0). The value of this flag determines what molecular species are included in the absorption of the Rayleigh scattered light that dominates the ultraviolet                                                                                                                                                                                                                                                                                                                                                                                                                                         |

spectrum. The value is a bitmap: for example, to include all the scattering species below, set IUVSCAT to 511 (the sum of all the bit values), or if you want include only C<sub>2</sub>H<sub>2</sub> and C<sub>2</sub>H<sub>6</sub> set IUVSCAT to 12 (the sum of the two bit values).

| Bit value | Species                       |
|-----------|-------------------------------|
| 1         | NH <sub>3</sub>               |
| 2         | PH <sub>3</sub>               |
| 4         | C <sub>2</sub> H <sub>2</sub> |
| 8         | C <sub>2</sub> H <sub>6</sub> |
| 16        | C <sub>4</sub> H <sub>2</sub> |
| 32        | CH <sub>4</sub>               |
| 64        | CO                            |
| 128       | GeH <sub>4</sub>              |
| 256       | H <sub>2</sub> O              |
| 512       | C <sub>3</sub> H <sub>8</sub> |

### 3.9 Additional reflecting atmosphere calculation definition .rfl file.

This file (<runname.rfl>), if present, contains the following lines:

- 1) Header (any)
- 2) Incident solar angle for calculation
- 3) Reflected zenith angle
- 4) Layer number of reflecting layer
- 5) Reflecting layer albedo

### 3.10 Additional vapour saturation definition .vpf file.

This file (<runname.vpf>), if present, lists the gases whose VMRs are to be limited by saturation and for each gas lists the desired limiting relative humidity and whether the volatile is arriving from the deep interior or from space. If the file is not present, then the vapour pressure of all gases is left untouched. Individual gas SVP curves are listed in the raddata/SVP.dat reference data file.

The first line of the file gives the number of gases, NVP, whose abundances are to be limited by condensation. There then flow NVP lines, each listing:

ID, ISO, VP, SVPFLAG

where ID and ISO are the identifiers of the gas concerned, VP is the limiting relative humidity required (normally between 0.0 and 1.0), and SVPFLAG is a control flag integer to govern the modification behaviour. SVPFLAG may take one of four values:

- 0 = ignore gas on this line (same as not having a line for this gas).
- 1 = apply the SVP-limited value at all levels.
- 2 = assume an interior source and disallow local minima allowed, i.e., the gas can only decrease with altitude
- 3 = assume an external source, but allow local minima for pressures less than 0.05 atm. The gas VMR can only decrease with decreasing altitude deeper than 0.05 atm. The value of 0.05 atm is what Nick Teanby using for Titan, but it should be OK for the giant planets too as any photochemical weirdness

is usually higher up.

At the moment, these flags do not cause any additional factors to be applied to XMAP, but this is something that we might want to think about depending on how well it works for different applications.

N.B. Technically XMAP should be set to 0.0 for gases whose VMRS are being limited by such condensation. However, we have found that this hard limit leads to undesirable retrieval behaviour such as: 1) if a gas VMR drops just below condensation in one iteration it can never return; and 2) it can lead to a sharp edge and erratic retrieval behavior. Therefore, to solve this XMAP is instead scaled by  $\text{svp/pp}$ . This also has a steep drop-off, but gives a more gentle response and more desirable retrieval behaviour.

### 3.11 Reduced wavelength scheme (.rdw) file

This file allows a Nemesis retrieval to be speeded up by giving each wavelength in a spectrum a 'ranking' (IRANK), and thereby allowing Nemesis to only retrieve a select sample of wavelengths in the first few iterations before moving on to the full wavelength grid. The procedure is as follows:

- In ITER=0, select only the wavelengths of IRANK=1 to perform the forward model (forwardnogX – B)
- Perform iterations with these IRANK=1 wavelengths until Nemesis converges to a solution as per the standard Nemesis convergence criteria
- Once convergence is achieved, add on the IRANK=2 wavelengths to the IRANK=1 wavelengths (forwardnogX – D), and perform further iterations in the same fashion.
- Repeat the same procedure until all wavelengths of  $\text{IRANK} \leq \text{MAXIRANK}$  are included. Once convergence is achieved with wavelengths of  $\text{IRANK} \leq \text{MAXIRANK}$ , the retrieval ends.
- If convergence is still not achieved after NITER-2 iterations, wavelengths of  $\text{IRANK} \leq \text{MAXIRANK}$  are added on automatically for the final 2 iterations.

The format of a typical .rdw file is as follows:

---

```
#Lines of comment preceded by a hash (or 'pound' if you're American) sign.
#Leave a blank line below the last comment
```

```
NCONV
ISPEC
MAXIRANK
ICONV(1) IRANK(1)
ICONV(2) IRANK(2)
...
ICONV(NCONV) IRANK(NCONV)
```

---

Note that the largest value of IRANK does not necessarily have to be smaller than MAXIRANK. It's just that any wavelengths with  $\text{IRANK} > \text{MAXIRANK}$  will be left out of the retrieval. This allows the user to easily remove any wavelengths out of the retrieval temporarily, without having to delete rows from the .spx file.

### 3.12 Non-LTE definition file (.lte)

Optional input file. Used to include some basic non-LTE emission effects. The **.lte** file (if present) is read in by `cirsrtfg_wave.f` and arguments are passed to `cirsradg_wave.f` using a common block. Approximations are implemented by modifying the emission temperature of the emitting layer in `cirsradg_wave.f` via `calc_nlte_t.f`. This is a pretty basic way to implement non-LTE so use with care! The functions are applied to **\_all\_** sources of opacity (ie all gases + continuum CIA + etc). This is only appropriate for a distinct emission line superimposed on an LTE continuum, assuming the continuum is generated at higher pressures where the non-LTE functions should tend to 1 (LTE) anyway. The functional form  $f=ap/(1+ap)$  applied to the Planck function ( $p$ =pressure(atm),  $a$ =constant) seems a good approximation to more detailed calculations, where a pressure of  $1/a$  corresponds to  $f=0.5$  and gives an idea of the pressures where non-LTE effects become important.

This first line of the .lte file is an integer flag.

Subsequent lines contain additional parameters. Possible flag values are:

| <b>nlte_flag</b> | <b>Approx</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                | No other parameters<br><br>Perfect LTE (ie do nothing)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1                | Second line contains $k_1$ and $A$<br><br>The Planck function is then scaled by this expression from Manuel Lopez Puertas:<br><br>$f = k_1 * N_{\text{tot}} / (A + k_1 * N_{\text{tot}})$ $N_{\text{tot}} = \text{total number density in molecules/cm}^3$ Using $PV = NkT$ gives:<br>$N_{\text{tot}} = N/V = p(\text{atm}) * 101325 * 1e-6 / (k_{\text{boltz}} * T)$ The temperature of the atmosphere is thus accounted for in the scaling.<br><br>Values of $k_1 \sim 10^{-12}$ are reasonable for most trace gas collisions at cold outer solar system temperatures. For example, for the non-LTE CH <sub>3</sub> emission on Titan at 16.5 microns reasonable values were:<br>$k_1 = 7.0 \times 10^{-13}$ $A = 3.23$ but $k_1$ has at least an order of magnitude uncertainty. See extended methods section of Nixon et al., (2025) for more details.<br><br>.lte file would contain here:<br>1<br>7.0e-13 3.23 |

|   |                                                                                                                                                                                                |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                                                                                                                                                                                                |
| 2 | <p>No other parameters.</p> <p>Orton approximation as in Radtrans manual (P27). A simple test function for Saturn/Titan. Planck function is scaled by:</p> $f = p * 1.0e6 / (1.0 + p * 1.0e6)$ |
|   |                                                                                                                                                                                                |
|   |                                                                                                                                                                                                |

## 4. Line-by-line calculations with Nemesis

### 4.1 Running Nemesis in LBL mode

**NEMESIS** now offers a LBL mode if ILBL is set to 1 in the .inp file. The set ups are identical to a normal correlated-k run, but additional input files are required to run **NEMESIS** in this mode:

1. A <runname.lbl> file is required. The first line of this file contains the wavenumber range (VMIN,VMAX) required to calculate the radiance over the wavelength/wavenumber range specified in the <runname>.spx file, together with the required wavenumber step for the LBL calculation. **NEMESIS** will calculate the LBL spectrum over this range with this step and then convolve with the instrument function specified by the <runname>.spx and <runname.sha> files. The code does not use an adaptive integrator and so the user needs to ensure that the calculation is performed at sufficient precision to model accurately the absorption features. The second line contains the LBL parameters: WING, VREL and VCUTOFF. These are described more fully in the accompanying Radtrans manual, but in summary the explicit line shape is used to calculate the contribution of a particular line to wavenumbers within WING. For larger distances from a line, it is assumed that we are into the Lorentz tail and so the contribution can be calculated at a much coarser resolution. VREL allows lines outside of VMIN,VMAX to contribute such that all the lines in the range VMIN-VREL to VMAX+VREL are included. Finally, VCUTOFF limits the contribution of any line past the specified distance from the line centres. This simulates the sub-Lorentzian behaviour of real lines. It is usual to set VREL=VCUTOFF. NB VREL will automatically be used to extend the wavenumber range defined by VMIN, VMAX. There is no need to pre-add/subtract VREL to the requested wavenumber range.
2. A <runname.sha> file is required to tell **NEMESIS** what instrument lineshape is required. The file contains a single line with a single integer: 0=square, 1=triangular, 2=Gaussian, 3=Hamming, 4=Hanning, 5=read from <runname>.fin file.
3. A <runname.fin> file if using ISHAPE=5 in <runname.sha>.sha file. First row is number of points (ninst). Subsequent ninst rows are two columns defining wavenumber/wavelength and lineshape.
4. A <runname.key> file is required which is a line data .key file as used for Radtrans LBL calculations, which specifies which line database is to be used and which gas information files. The contents of this file are described in the Radtrans manual.

5. Optionally a <runname.pra> can be provided. If present, this file lists which line shape should be used for particular gases. If the file is absent then the Voigt lineshape is used for all gases. If present, the file contains one row for each gas to be modified containing 'ID ISO IPROC', where ID, ISO are Radtrans ID/Isotope numbers of the gas to be modified and IPROC is required lineshape. Allowed values of IPROC are listed in the Radtrans manual.

## 4.2 Running Nemesis in pretabulated LBL mode.

**NEMESIS** now offers a pre-tabulated LBL mode if ILBL is set to 2 in the .inp file. The set ups are identical to a normal correlated-k run, but additional input files are required to run **NEMESIS** in this mode:

1. A <runname.sha> file is required to tell **NEMESIS** what instrument lineshape is required. The file contains a single line with a single integer: 0=square, 1=triangular, 2=Gaussian, 3=Hamming, 4=Hanning.
2. A <runname.lls> file is required, which fulfills the same function as the <runname.kls> file for a correlated-k calculation. The <runname.kls> lists the pre-tabulated monochromatic gas absorption lbl-tables (pre-tabulated with Calc\_lbltable, rather than Calc\_fnktablec as for k-tables), which like .kta files are in direct-access binary format and have a .lta file extension.

## 5. Location of code and example input files

**NEMESIS** source code is now under Github management to maintain all versions of the software that might be released. The Github central repository of **RADTRANS** (and **NEMESIS**) is <https://github.com/patirwin123/radtrancode> (please contact P. Irwin for details on how to access the files). Please see the Radtrans manual, section 1.1, for further details.

## 6. Recent Developments

Since the last version of **NEMESIS**, the following notable changes have been made:

1. The code can now do LBL calculations.
2. The code can deal with profiles where the sum of vmrs adds up to 1 and so the molecular weight can be calculated at each level.

## 7. Future developments

**NEMESIS** is intended to be the main retrieval tool of the Oxford Planetary Data Analysis group for future missions and is designed to be general purpose and extendable. Future upgrades that are under consideration and may/may not be implemented in the near future are:

1. Update to allow 'gradient' or implicit differentiation to determine Jacobian for scattering retrievals (hard). This has already been attempted, but the resulting code was actually slower than the numerical differentiation scheme. To implement this in a way that gains any advantage may require some clever and elegant reprogramming.

The overarching intention of **NEMESIS** is to provide a single retrieval code that can be applied to any planet and in which improvements and debugs made in one research application are then available to researchers analysing different data. While every attempt has been made to adhere to this goal, some cases have arisen where it has proven necessary to form an offshoot. These versions are different from the central **NEMESIS** version on the way they set up the spectral calculation and the way the underlying radiative transfer calculations are combined to give the final result. Specifically the different offshoots generate different .pat files, which in turn generates different .drv files and the outputting path calculations are recombined differently. The current offshoots are as follows.

NemesisL is specifically designed for limb calculations. The atmosphere is split into the same number of layers, NLAYER, as before, but only once and from the specified lowest altitude. Limb paths are then calculated through these layers with  $2 \times \text{NLAYER}$  layers included in the lowest path and 2 layers in the top path.

ISPACE, ISCAT, OCCULT, ILBL, INUM

The transmission and/or thermal emission for these tangent paths are calculated once and then interpolated to the actual tangent altitude required. The OCCULT parameter controls what sort of limb calculation is required. The options are:

OCCULT = 0. Thermal emission calculation only  
OCCULT = 1. Solar irradiance\*transmission + thermal emission  
OCCULT = 2. Transmission calculation only  
OCCULT = 3. (Solar irradiance\*transmission + thermal emission)/  
(Solar irradiance)

For most cases the Jacobians can be calculated analytically, for which INUM=0. However, when trying to retrieve temperature and pressure together, the effects of varying each changes the atmospheric profile in a complicated and non-linear way. Hence, these Jacobians have to be calculated by numerical differencing, set by INUM=1.

For a single limb calculation, you should use Nemesis, but for multiple tangent heights through the same atmosphere, NemesisL is much faster and not significantly less accurate.

NemesisMCS is based on NemesisL, but is specifically tailored to model MCS radiances, which require a complicated FOV combination of the individual path calculations. Other modifications are made in the way the spx files are read in, and implicitly use small differences in the wavenumbers listed to identify different detectors, which have slightly different spectral and FOV responses.

### 8.3 Nemesisdisc

Nemesisdisc sets up the .pat file for a disc-averaged calculation and outputs the results in a form compatible with modelling the secondary transits of exoplanets.

### 8.4 NemesisPT

NemesisPT is again based on NemesisL, but instead uses the limb path calculations to estimate the effective planetary radius at different wavelengths. This code is thus used for modelling the primary transits of exoplanets.

### 8.5 CIRSdrv\_wave

CIRSdrv\_wave is a very useful program for running the nemesis/cirsrad subroutines on cases that are not covered by the Nemesis programs, which write their own .pat files to model specific geometries. CIRSdrv\_wave can be run in command line, but you're bombarded with quite a few questions and so I often run it as:

```
CIRSdrv_wave < cirsdrv_wave.inp > test.prc &
```

A typical cirsdrv\_wave.inp file would look like this:

|            |                                                                 |
|------------|-----------------------------------------------------------------|
| zero       | (root name of the .pat, .fla, .kls and .cia files)              |
| 1          | (iwave: 0=wavenumber, 1=wavelength)                             |
| y          | (Do you want the wavelengths extracted from the .pat file?      |
| 3          | (IPLANET, 3=Earth, 4=Mars, etc.)                                |
| 1          | (1=Fast, 2=slow. Historical – always choose '1')                |
| 0.35 0.005 | (minimum wavelength/wavenumber and step of k-tables used)       |
| 11         | (This defines which scloudNNwave.f model to use, if scattering) |
| 200.       | (Surface temperature)                                           |

The same inputs are always required even when they're not actually necessary! Hence, you need to provide a number for scloudNNwave and surface temperature even for calculations where these numbers are irrelevant. The basic advantage of CIRSdrv\_wave is that it can run on ANY .pat file and so can test situations outside of the usual Nemesis-like calculations. For example, you can use it to calculate transmission weighting functions. The output of CIRSdrv\_wave is an ASCII file containing the computed spectra for each path defined in the .pat file. Two spectra are output per path, one which is the raw spectrum calculated at the k-table wavelengths/wavenumbers spanning the requested range and the second is this spectrum convolved with a box of width FWHM (defined in the .pat file) and output at the wavelengths/wavenumbers specified in the .pat file.

### 8.6 Lbldrv\_wave

Lbldrv\_wave is exactly the same as CIRSdrv\_wave except that it uses the new nemesis/cirsrad LBL subroutines. The input script is essentially identical, except that there is now no need to define the k-table minimum wavelength/wavenumber and step. However, the user does need to provide the additional .key, .lbl and .sha files necessary for a nemesis-LBL run. The output is also an ASCII file, but here there is one spectrum per defined path, smoothed with the FWHM defined in the .pat file and with the instrument function defined in the .sha file.

## 9. Notes

### 9.1 Matrix inversion instability

Debugging **NEMESIS** it was noticed that the cost function, calculated by `calc_phiret.f` was sometimes returning negative numbers. Closer investigation revealed that the problem arose from inverting the *a priori* covariance matrix, which because of the way the correlation length was used contained loads of off-diagonal elements which were almost zero but not quite. This led to instabilities in calculating the inverse of the matrix which could not be rectified even by going to double precision. **NEMESIS** now sets off-diagonal elements using the usual correlation-length formula used by **Oxcirs** and **Oxcirsg** except that off-diagonal elements smaller than a certain prescribed factor are now set to zero. This modification makes the matrix inversion stable and leads to sensible cost function values. The retrievals of **NEMESIS** now converge smoothly. The *a priori* covariance matrix is then inverted in double precision with a Cholesky decomposition routine that checks to ensure the resultant inverse actually works (i.e.,  $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ ). The inverse is then stored to speed up resultant routines.

N.B. The same ‘feature’ is probably also present in **Oxcirs** and **Oxcirsg** and probably explains why during the retrievals, the Marquardt-Levenburg parameter **ALAMBDA** sometimes reduces nicely until the `calc_phiret.f` deduces that the solution starts getting further from the optimal fit. **ALAMBDA** then increases at every iteration until the maximum iteration number is reached. This behaviour can be explained if `calc_phiret.f` is calculating slightly the wrong value of the cost function and thus while the retrieval routine is trying to get closer to the solution, the cost function routine thinks it is getting further away! This means that in some previous retrievals, the presented solutions were probably not quite converged if this behaviour was present.

### 9.2 Constraints and ‘exact’ solutions

During retrieval tests of  $\text{PH}_3$  and  $\text{NH}_3$  profiles in Saturn’s atmosphere it became clear that the retrieved solution was weighted too strongly by the measurements and tended to the ‘exact’ solution. In this case the correlation length set in the *a priori* profiles has very little effect and the solution becomes ‘wiggly’ and unattractive. I found it initially rather difficult to judge how to better constrain the solution and how to test if it was constrained!

To investigate this I went back to look at the Barney Conrath approach (see chapter 8 in [R3]) and found many similarities with optimal estimation, but perhaps a more realistic way of considering the constraints. In the optimal estimation approach used, the solution (in the non-linear case) is:

$$\mathbf{x}_{n+1} = \mathbf{x}_0 + \mathbf{S}_x \mathbf{K}_n^T (\mathbf{K}_n \mathbf{S}_x \mathbf{K}_n^T + \mathbf{S}_\varepsilon)^{-1} (\mathbf{y}_m - \mathbf{y}_n - \mathbf{K}_n (\mathbf{x}_0 - \mathbf{x}_n)) \quad (8.1)$$

where  $\mathbf{S}_x$  and  $\mathbf{S}_\varepsilon$  are the *a priori* and measurement covariance matrices respectively and  $\mathbf{K}_n$  is the matrix of functional derivatives, or Jacobian.

In Conrath’s retrieval method (as I understand it!) the formalism is slightly different:

$$\mathbf{x}_{n+1} = \mathbf{x}_0 + \hat{\mathbf{S}}_x \mathbf{K}_n^T \left( \mathbf{K}_n \hat{\mathbf{S}}_x \mathbf{K}_n^T + \gamma \mathbf{S}_\varepsilon \right)^{-1} (\mathbf{y}_m - \mathbf{y}_n - \mathbf{K}_n (\mathbf{x}_0 - \mathbf{x}_n)) \quad (8.2)$$

where  $\hat{\mathbf{S}}_x$  is now the *a priori* correlation matrix,  $\mathbf{S}_\varepsilon$  and  $\mathbf{K}_n$  are as before, and  $\gamma$  is an adjustable parameter to fine-tune the balance between measurement and *a priori* constraint.

Clearly the two formalisms are extremely similar with, I think, the optimal estimation approach more flexible in that different elements of the measurement vector can have different *a priori* constraints whereas they are all the same in Conrath's approach (if their correlation matrix is what I think it is, i.e.,  $\hat{S}_{ij} = S_{ij} / (S_{ii} \times S_{jj})^{1/2}$ ).

I find it very difficult to decode some of the work of Clive Rodgers [R4], excellent though it is. In Rodgers (2000) there are pages and pages discussing constraints, contribution functions, averaging kernels and error propagations without ever (as far as I can tell) arriving at a simple way of judging whether there is enough constraint in the retrieval or whether it tends to the 'exact' and thus unsmoothed and unreliable one.

I have now thought of a simple test! In Eq. 8.1 if the measurement errors are tiny, the contribution function, or gain matrix,  $\mathbf{G}_n$  tends to:

$$\mathbf{G}_n = \mathbf{S}_x \mathbf{K}_n^T (\mathbf{K}_n \mathbf{S}_x \mathbf{K}_n^T + \mathbf{S}_\varepsilon)^{-1} \rightarrow \mathbf{S}_x \mathbf{K}_n^T (\mathbf{K}_n \mathbf{S}_x \mathbf{K}_n^T)^{-1}$$

and the solution is exact. If however the measurement errors are huge then the contribution function  $\mathbf{G}_n$  tends to:

$$\mathbf{G}_n = \mathbf{S}_x \mathbf{K}_n^T (\mathbf{K}_n \mathbf{S}_x \mathbf{K}_n^T + \mathbf{S}_\varepsilon)^{-1} \rightarrow \mathbf{S}_x \mathbf{K}_n^T \mathbf{S}_\varepsilon^{-1}$$

and too much constraint is applied.

The happy medium seems to me to be when the diagonal elements of  $\mathbf{K}_n \mathbf{S}_x \mathbf{K}_n^T$  are of a similar magnitude to the diagonal elements of  $\mathbf{S}_\varepsilon$  (which is usually diagonal anyway). This is now tested for in the code and reported. Errors in either the *a priori*, or the measurement covariance matrices can then be adjusted to ensure sufficient constraint and this sufficient vertical smoothing. This may be achieved by modifying the extra forward modelling error file (section 3.2) and is very much akin to modifying the  $\gamma$  trade-off parameter in Conrath's approach. Alternatively, by reducing the *a priori* errors or increasing the correlation length, the magnitude of  $\mathbf{K}_n \mathbf{S}_x \mathbf{K}_n^T$  can be reduced leading to a similar result.

There are of course other ways of assessing the retrieval stability and the IDL code `imagecovariance.pro` which reads the <runname>.cov file (section 2.3) displays every diagnostic plot imaginable which should help to investigate any retrieval problems.

Optimal estimation was developed for terrestrial retrievals where there is good statistical knowledge of the expected state vectors. This is NOT the case for planetary work where what our method (and that of Barney Conrath's) actually does is to extract a smoothed representation of a real continuous profile from a limited set of data. For a reliable retrieval we require that the level of smoothing is sufficient to damp the ripples

that appear in exact retrievals. If the *a priori* errors are large, then the retrieval is exact and although the retrieved errors are small, the profile is ‘wiggly’. As the *a priori* errors are reduced, more and more smoothing is applied and so the retrieval errors reduce further but the retrieved profile is smoother. Another test for sufficient constraint may be to compare the measurement error  $S_m$  and the smoothing error  $S_n$ . These would appear to have to be of similar size at the peak of the weighting functions when the retrieval is ‘balanced’. This is still being investigated.

### 9.3 Optimisation of retrieval code.

The equation we solve in these retrievals is:

$$\mathbf{x}_{n+1} = \mathbf{x}_0 + \mathbf{S}_x \mathbf{K}_n^T (\mathbf{K}_n \mathbf{S}_x \mathbf{K}_n^T + \mathbf{S}_\varepsilon)^{-1} (\mathbf{y}_m - \mathbf{y}_n - \mathbf{K}_n (\mathbf{x}_0 - \mathbf{x}_n)) \quad (8.3)$$

or equivalently

$$\mathbf{x}_{n+1} = \mathbf{x}_0 + \mathbf{G}_n (\mathbf{y}_m - \mathbf{y}_n) - \mathbf{A}_n (\mathbf{x}_0 - \mathbf{x}_n) \quad (8.4)$$

where  $\mathbf{G}_n$  is the gain matrix and  $\mathbf{A}_n (= \mathbf{G}_n \mathbf{K}_n)$  is the averaging kernel matrix. Previous versions of the code calculated the gain matrix as:

$$\mathbf{G}_n = \mathbf{S}_x \mathbf{K}_n^T (\mathbf{K}_n \mathbf{S}_x \mathbf{K}_n^T + \mathbf{S}_\varepsilon)^{-1} \quad (8.5)$$

which for cases where the length  $m$  of the measurement vector was large, tended to be extremely slow. However, an equivalent formulation of the gain matrix is:

$$\mathbf{G}_n = (\mathbf{K}_n^T \mathbf{S}_\varepsilon^{-1} \mathbf{K}_n + \mathbf{S}_x^{-1})^{-1} \mathbf{K}_n^T \mathbf{S}_\varepsilon^{-1} \quad (8.6)$$

Since  $\mathbf{S}_\varepsilon$  is assumed to be diagonal, and since  $\mathbf{S}_x^{-1}$  is pre-computed in order to work out the cost function, this formulation is much faster to calculate for  $n < m$ .

The gain matrix (and the averaging kernel matrix) are now calculated by the subroutine `calc_gain_matrix.f`, and the calculation of Eq. 8.4 done by `calcnexn.f`. These two routines replace the old `dretrieve.f` subroutine. Having the gain matrix and averaging kernel matrix available also makes the calculation of the final errors in `calc_serr.f` much faster too! One final improvement is that Cholesky, rather than LU decomposition, is now used for the matrix inversion, which is an optimised code for inverting positive-definite matrices, such as covariance matrices.

In version A7 onwards, the results of previous retrievals may be included (together with error) by incorporating them, appropriately, into  $\mathbf{S}_\varepsilon$ .

### 9.4 Adding New Parameterisations

If you want to add a new profile parameterisation to **NEMESIS**, then there are numerous steps to complete:

1. You need to add another parameterisation definition in `readapriori.f`. In `readapriori.f` you need to decide what new parameterisation scheme you want and which of the state vector elements need to be held internally as log values and which are held as linear values. For example, we always hold volume mixing ratios as log values so that the actual vmrs that get written to

the .prf file can never become negative. For each variable type, there are three variables initially read in: VARIDENT(IVAR,1-3), which defines what sort of profile this is. This is all defined in section 3.1, but in summary:

- a. If VARIDENT(IVAR,1)=0, then this is a temperature profile
  - b. If VARIDENT(IVAR,1) > 0 and < 100, then this defines a gas volume mixing ratio profile, in which case IDGAS = VARIDENT(IVAR,1) and ISOGAS = VARIDENT(IVAR,2).
  - c. If VARIDENT(IVAR,1) is negative then the profile defines either an aerosol abundance profile, a para-H<sub>2</sub> fraction profile, or fractional cloud cover. See section 3.1 for details.
  - d. If VARIDENT(IVAR,1) < 100, then VARIDENT(IVAR,3) defines the profile type. If you have a new profile parameterisation in mind, then please pick the next unused number.
  - e. If VARIDENT(IVAR,1) > 100, then this defines a ‘special’ parameterisation that might use a special layering technique, or define the cloud imaginary refractive index spectrum, etc. In such cases, please set VARIDENT(IVAR,3) = VARIDENT(IVAR,1), and set VARIDENT(IVAR,2) = 0.
2. You also need to update `logflag.f` to correctly identify which of the new state vector elements are held as log values.
  3. If you are defining a new temperature, gas volume mixing ratio, cloud, para-H<sub>2</sub> or fractional cloud cover profile parameterisation you also need to update `npvar.f`, which defines how many elements in the state vector, NP, are necessary to define the new parameterisation.
  4. If you are defining a ‘special’ parameterisation scheme then you will have a considerably more painful time as you need to define NP (the number of variables associated with each parameterisation in the state vector) individually in the following files: `check_iteration.f`, `coreret.f`, `coreretdisc.f`, `coreretMC.f`, `extractsrom.f`, `extractsromch4.f`, `generateMCSpfx.f`, `generatespfx.f`, `gsetrad.f`, `modxvecMCMC.f`, `modxcevMCMCA.f`, `readraw.f`, `rmodapriori.f`, `scankkx.f`, `setifix.f`, `writeout.f`. This is not very elegant and should probably be improved. However, you will hopefully just be making a new temperature, gas volume mixing ratio, cloud, para-H<sub>2</sub> or fractional cloud cover profile parameterisation and so will not need to worry about this step!
  5. You need to edit `subprofretg.f` to use the new x-vector elements to modify either the <runname>.prf, aerosol.prf, parah2.prf, or fcloud.prf files. You also need to define the internal gradients, which is a little trickier. For each profile parameterisation, `subprofretg.f` takes the NP state vector elements and constructs the profile as you require, but you also need to define the relative elements of the XMAP(MAXV,MAXGAS+2+MAXCON,MAXPRO) array. XMAP is a functional derivatives matrix and elements of XMAP are the rate of change of the profile vectors (i.e., temperature, vmr, etc.) with respect to the rate of change in the state vector elements. So, for example, if X1(J) is the modified temperature, vmr, clouds at level J to be written out to runname.prf or aerosol.prf then XMAP(K,L,J) is  $d(X1(J))/d(XN(K))$ , where L is the internal variable identifier which identifies what kind of profile this is and lies in the range (1 to NGAS+1+2\*NCONT). The different values of L are:

- a. If L is in the range (1,NGAS), then we're defining a gas volume mixing ratio.
- b. If L = NGAS+1, then this is a temperature profile.
- c. If L is in the range (NGAS+2,NGAS+2+2\*NCNT), then this is an aerosol profile, para-H<sub>2</sub> fraction or fractional cloud cover profile.

You need to think carefully about how the gradients are calculated. You also need to think about whether the individual elements in the state vector are held as linear or log values. Obviously if  $X1(J) = XN(K)$ , then  $d(X1(J))/d(XN(K)) = 1$ , but if  $X1(J) = \exp(XN(K))$ , then  $d(X1(J))/d(XN(K)) = \exp(XN(K))$ . The best thing to do is find an existing profile parameterization that most closely matches what you want to do, study how it and the XMAP elements are defined and then define a new profile type based on this and modify what you need to change. That's what I always do!

## 10. K-table location

To facilitate future research, attempts are ongoing to co-locate k-tables generated for various projects by the Nemesis modelling community. This k-tables are not currently part of the Nemesis/Radtrans distribution, but for Oxford users, the k-tables may be found in: /network/group/aopp/planetary/PGJI001\_IRWIN\_LBLKTAB.

## 11. Note on k-table formats

By default, it is assumed that k-tables are compiled with square averaging bins of width FWHM, with bin centres spaced by DELV, starting at VMIN. Normally,  $DELV = FWHM/2$  to achieve Nyquist-sampling. If  $DELV > 0$ , then **NEMESIS** works which wavelengths/wavenumbers in the table cover the range of interest defined in the <runname>.spx file and then convolves the output with a second box of full-width-half-maximum, FWHM1, defined in the <runname>.spx file. Usually,  $FWHM1 = FWHM$  and so the resulting convolution of two square boxes gives a triangular instrument lineshape (ILS) with the same full-width-half-maximum. If FWHM1 is set to zero, then the output spectrum is simply interpolated to the required wavelengths/wavenumbers and has a square ILS of full-width-half-maximum equal to that of the k-table, i.e., FWHM.

If  $DELV > 0$  and  $FWHM=0$  in the k-table, then it is assumed that a non-square lineshape has been used to compile the k-table over a regular grid of wavelengths/wavenumbers. This will not normally require further convolution and so FWHM1 should be set to zero in the <runname>.spx file and the output is interpolated to the required wavelengths/wavenumbers.

If  $DELV < 0$  then the k-table has been calculated for a non-regular set of central wavelengths/wavenumbers, with, usually, non-square lineshapes that have a FWHM that is different for every central wavelength or wavenumber in the table. In other words, the k-table describes a set of filter-averaged channels. In such tables, FWHM is set by convention to be equal to zero. **NEMESIS** will compare the requested wavelengths in the <runname>.spx file, with those available in the k-table and 'snap' to the nearest available entry to run the calculation. No further interpolation or smoothing is applied. This method is the fastest calculation as only one entry in the k-table is used for each output wavelength/wavenumber in the <runname>.spx file and

no further convolution is applied. It is possible for such tables to be regularly gridded and have the same ILS at all wavelengths.